

A Comprehensive Study on Modelling and Control of Autonomous Underwater Vehicle

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Abstract. Autonomous underwater vehicles (AUV) have become the de facto vehicle for remote operations involving oceanography, inspection, and monitoring tasks. These vehicles operate in different and often challenging environments; hence, the design and development of the AUV involving hydrodynamics and control systems need to be designed in detail. This book chapter presents a study on the modelling and robust control of a research vehicle in the presence of uncertainties. The vehicle's dynamic behaviour is modelled using a 6-degree-of-freedom approach, considering the effect of ocean currents. The level flight requirements for different speeds are derived, and the resulting model is decomposed into horizontal and vertical subsystems for linear analysis.

The simulation results presented focus on the efficacy of linear controllers within three key subsystems: depth, yaw, and speed. Moreover, level-flight outcomes are demonstrated for a speed of 4 knots. The nonlinear control strategies employed in this study encompass conventional and sliding-mode control (SMC) methodologies. To ensure accurate tracking performance, the controller design considers the vehicle's dynamics with various uncertainties such as ocean currents, parameter uncertainty, CG (Center of Gravity) deviation and buoyancy variation. Both conventional and nonlinear SMC controllers' outcomes are showcased with a lawn-mowing manoeuvre scenario. A systematic comparison is drawn between the robustness of SMC against disturbances and parameter fluctuations in contrast to conventional controllers. Importantly, these results underscore the trade-off that accompanies SMC's robustness, as it necessitates a higher level of complexity in terms of controller design, intricate implementation intricacies, and the management of chattering phenomena.

Keywords: AUV, Modelling, Ocean currents, Level-flight/Trim, Control architecture, Parameter uncertainties, Sliding mode control,

1 Introduction

Vastness and inaccessibility of the ocean have led to several challenges in underwater exploration and research. There has been a growing interest in developing AUVs to achieve better exploration and data collection in the last few years. The AUVs are unmanned vehicles that are capable of operating on their own in underwater environments without direct human intervention. The development of these AUVs has opened several fields, especially in marine technology, marine biology, geology and oceanography. Because of their ability to navigate to difficult underwater terrains, and collect high-resolution scientific data, they have a key role in seafloor mapping, aircraft crash investigations, marine rescue, military applications and Intrusion surveillance.

Depending on the specific application or operating condition there are different sizes, shapes and configurations of AUVs. Also, these applications require the AUV to carry different sensors like cameras, sonar, depth sensors, navigational sensors or some other scientific instrument. Although

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Abstract. 自主水下航行器 (AUV) 已经成为涉及海洋学、检测和监测任务的远程操作的事实标准车辆。这些车辆在不同且常常具有挑战性的环境中运行; 因此, 涉及流体动力学和控制系统的 AUV 设计与开发需要精细设计。本书章节介绍了一项关于在存在不确定性情况下研究型车辆建模与鲁棒控制的研究。车辆的动态行为采用六自由度方法建模, 并考虑海流的影响。不同速度下的平飞要求被推导出来, 所得模型被分解为用于线性分析的水平和垂直子系统。

所呈现的仿真结果主要关注于线性控制器在三个关键子系统 (深度、偏航和速度) 中的有效性。此外, 还展示了以4节速度进行平飞的结果。本研究中采用的非线性控制策略包括传统控制和滑模控制 (SMC) 方法。为了确保精确的跟踪性能, 控制器设计考虑了车辆动态及各种不确定性因素, 如洋流、参数不确定性、重心 (CG) 偏移和浮力变化。采用割草机机动场景展示了传统控制器和非线性 SMC 控制器的结果。系统地比较了 SMC 在应对干扰和参数波动方面的鲁棒性与传统控制器的差异。重要的是, 这些结果凸显了 SMC 鲁棒性所伴随的权衡, 因为这需要在控制器设计和复杂性上付出更高的代价。

Keywords: 自主水下航行器, 建模, 洋流, 平飞/修剪, 控制架构, 参数不确定性, 滑模控制,

1 Introduction

海洋的广阔性和难以接近性导致了水下探索和研究的诸多挑战。近年来, 人们对开发自主水下航行器 (AUV) 的兴趣日益增加, 以实现更好的探索和数据收集。AUV 是能够在水下环境中自主操作, 无需直接人为干预的无人驾驶车辆。这些 AUV 的发展开辟了多个领域, 特别是在海洋技术、海洋生物学、地质学和海洋学方面。由于其能够导航到复杂的水下地形并收集高分辨率科学数据, 它们在海底测绘、飞机事故调查、海洋救援、军事应用和入侵监控中发挥着关键作用。

根据具体的应用或操作条件, AUV 有不同的尺寸、形状和配置。此外, 这些应用要求 AUV 携带不同的传感器, 如摄像机、声纳、深度传感器、导航传感器或其他科学仪器。尽管如此

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these developments and capabilities are fascinating, the underwater environment is characterized by its harsh conditions because of which there are several challenges that needs to be addressed. Due to the constraints of underwater wireless communication, AUVs have drawn more attention from the underwater scientific community. Autonomous decision-making underwater robots are becoming more dependable and beneficial due to improvements in AUV research, materials, manufacturing procedures, sensors, processing power, and battery technology. The goal of current AUV research is to develop a trustworthy, completely autonomous robotic decision-making system.

A "Fish"-like torpedo created by Robert Whitehead in 1886 is considered to be the first AUV in history [1]. The first torpedo had a range of 700 metres, was powered by compressed air, and had a top speed of 3.0 m/s. In 1957, Special Purpose Underwater Research Vehicle (SPURV) was developed by the Applied Physics Laboratory at the University of Washington. The vehicle can dive to a depth of 3650 m and travel at a speed of 2-2.5 m/s [2]. The goal was to collect oceanographic data using precise trajectory calculations. More than 400 deployments of SPURV were documented up until the late 1970s to carry out diverse duties before SPURV-II was built in 1973 [1].

In the last two decades, several AUVs have been developed by various organizations across the globe. Some noteworthy AUVs are namely Remote Environmental Monitoring UnitS (REMUS) and its variants [1], [3], [4] and [5], Autosub 6000 [6], Indian origin AUVs MAYA [7], [8] and AUV 150 [9], LAUV [10], Odyssey AUV series [11], Bluefin12 and Bluefin21 [12] -[14], HUGIN HUS AUV [15]. These AUVs are employed for surveillance, mine countermeasures, search and rescue, bathymetry, mapping the ocean floor, geological survey, environmental monitoring, repair and maintenance, structure inspection and several other underwater applications. Applications and technological challenges are tabulated in Table 1.

Table 1: AUV Applications

	Application	Technological challenges
Military	Surveillance	Localization, path planning control, guidance, obstacle avoidance, trajectory tracking
	Mine Countermeasures	
	Target localization	
	Search and rescue	
	Ocean exploration	
	Anti-submarine warfare	
Scientific	Geological survey	Obstacle avoidance, path planning, trajectory tracking, control
	Marine biology study	
	Archaeological survey	
	Environmental monitoring	
	Seafloor mapping	
Industry	Gas and oil inspection	Localization, control, guidance
	Pipeline and infrastructure maintenance	
Commercial	Underwater video footage collection	Control, obstacle avoidance
	Entertainment and Tourism	
	Fishing	

这些发展和能力令人着迷，水下环境因其恶劣的条件而具有独特的特点，因此存在需要解决的若干挑战。由于水下无线通信的限制，自动水下航行器（AUVs）引起了水下科学界更多的关注。随着AUV研究、材料、制造工艺、传感器、处理能力和电池技术的改进，水下自主决策机器人正变得更加可靠和有益。当前AUV研究的目标是开发一个值得信赖的、完全自主的机器人决策系统。

一种由罗伯特·怀特黑德在1886年创造的“鱼”形鱼雷被认为是历史上第一种自主水下航行器（AUV）[1]。第一枚鱼雷的射程为700米，由压缩空气驱动，最高速度为3.0米/秒。1957年，华盛顿大学应用物理实验室开发了特种水下研究艇（SPURV）。该车辆可以下潜至3650米深，并以2-2.5米/秒的速度行驶[2]。其目标是通过精确的轨迹计算收集海洋数据。截至1970年代末，SPURV已进行超过400次部署，以执行各类任务，直到1973年建造了SPURV-II[1]。

在过去的二十年中，全球各地的多个组织开发了几种自主水下航行器（AUV）。一些值得注意的AUV包括远程环境监测单元（REMUS）及其变体 [1], [3], [4] 和 [5], Autosub 6000 [6], 印度原产的AUVs MAYA [7], [8] 和 AUV 150 [9], LAUV [10], Odyssey AUV系列 [11], Bluefin12和Bluefin21 [12]-[14], HUGIN HUS AUV [15]。这些AUV被用于监视、排雷、搜救、测深、海底测绘、地质调查、环境监测、维修及维护、结构检查及其他多种水下应用。应用与技术挑战列在表1中。

表 1: AUV 应用

	Application	Technological challenges
Military	Surveillance	Localization, path planning control, guidance, obstacle avoidance, trajectory tracking
	Mine Countermeasures	
	Target localization	
	Search and rescue	
	Ocean exploration	
	Anti-submarine warfare	
Scientific	Geological survey	Obstacle avoidance, path planning, trajectory tracking, control
	Marine biology study	
	Archaeological survey	
	Environmental monitoring	
	Seafloor mapping	
Industry	Gas and oil inspection	Localization, control, guidance
	Pipeline and infrastructure maintenance	
Commercial	Underwater video footage collection	Control, obstacle avoidance
	Entertainment and Tourism	
	Fishing	

The successful operation of the AUVs depends on the mathematical model and control architecture. They play a pivotal role in enhancing the vehicle capabilities and overall performance. The challenges AUVs face in underwater environments, such as complex hydrodynamics, uncertain external forces, and the need for precise manoeuvring, necessitates the development of sophisticated control strategies for stability, accuracy, and mission success. Over the past two decades, the research community has shown interest in employing neural networks for system identification and control, marking a significant advancement in the field. Pioneering work by Narendra [16] set the stage, followed by a sequence of endeavours that utilized actual vehicle data for system identification [17] - [24]. Within the realm of adaptive neural network models, diverse architectures emerged, encompassing nonlinear neuroadaptive methods [17], [18], [19], recurrent neural network [20], neural networks with augmented model inversion [21], and the recent robust EMRAN approach for autonomous vehicles [24]. While predominantly explored in aerial vehicles necessitating agile manoeuvres, these technologies are progressively making their mark in underwater vehicles. This transition extends to AUVs, where analogous identification methodologies have been presented for navigation [25]. Further advancements in underwater vehicle speed control have leveraged self-propulsion estimation models [26], enabling the prediction of propeller rotation rates for a targeted speed. Notably, advancements in aggressive manoeuvres and redundant actuators, exemplified by technologies such as [27], are finding resonance in the field, poised to play a pivotal role in underwater vehicle systems.

In spite of these advancements, most of these underwater vehicles are slow-moving and less agile in manoeuvring in nature. Moreover, one of the major challenges in the underwater environment is the disturbance due to the ocean current or the turbulence due to the motion of the large obstacles near the operation of the underwater vehicles. Hence most of the research focuses on conventional reliable robust control methods. Over the years, numerous control architectures have been developed for AUVs that offer a wide array of controller options. Comprehensive survey articles and related references [28] - [32] provides an extensive overview of the diverse controllers used for AUVs. From the early stages of AUV experimentation, PID controllers emerged as a simple and popular choice [33] - [36]. These controllers were employed for various AUV subsystems, including speed control, depth control, and steering control, often using linear PID designs [37], [38], [39]. For instance, the REMUS AUV utilizes a set point PD controller with regulator gain matrices to drive linear position and orientation errors to zero [37]. Further, fuzzy self-adapting PID controllers [40] and LMI-based PID controllers for wave disturbance suppression [41] have been presented. In [42], an auto-tuned PID-like controller based on neural networks is designed. The PID controller's weights are tuned online to minimize position-tracking errors. To address parameter uncertainties, a fractional order PID controller has been designed for yaw control [43], offering robustness against variations of up to $\pm 30\%$ from the actual parameters.

Disturbance and perturbation are the main concerns in underwater vehicles. This affects the control performance of the system. A reliable method that is frequently applied to underwater vehicle control to provide robustness is Sliding Mode Control (SMC). This method provides robustness against bounded external perturbations and ensures finite-time convergence. Sliding mode control (SMC) has been widely used in literature [44] - [47]. In early implementations, SMC was tested on a nonlinear underwater vehicle model using three different second-order controllers for position and velocity, demonstrating its effectiveness [44]. A modified SMC is implemented in [48] to deal with changing dynamics and operating conditions in the dive plane. Authors in [49] present a robust adaptive sliding mode control law with a velocity state observer for a 3 DOF AUV model, highlighting its capability to handle disturbances and uncertainties in parameters. An SMC controller is designed for a 6-DoF model nonlinear system [50]. The robustness of SMC is demonstrated with the simulation results. Furthermore, Decoupled systems, such as horizontal and vertical planes, are commonly used in AUVs with SMC, making them robust against bounded disturbances as shown in [51].

AUV的成功运行依赖于数学模型和控制架构。它们在增强车辆能力和整体性能中起着关键作用。AUV在水下环境中面临的挑战，如复杂的水动力学、不确定的外力以及对精确机动的需求，迫切需要开发复杂的控制策略，以确保稳定性、精确性和任务成功。在过去的二十年中，研究界对利用神经网络进行系统识别和控制表现出浓厚兴趣，这标志着该领域的重大进展。Narendra [16] 的开创性工作奠定了基础，随后一系列利用实际车辆数据进行系统识别的努力出现 [17]-[24]。在自适应神经网络模型领域，出现了多种架构，包括非线性神经自适应方法 [17], [18], [19]、递归神经网络 [20]、增广神经网络

尽管取得了这些进展，大多数水下车辆在移动速度和机动性方面仍然较慢。此外，水下环境的一个主要挑战是由于海洋洋流造成的扰动，或者由于大型障碍物运动引起的湍流，从而影响水下车辆的操作。因此，大部分研究集中在传统的、可靠的稳健控制方法上。多年来，为AUV开发了大量控制架构，提供了广泛的控制器选择。全面的综述文章和相关参考文献 [28] - [32] 对用于AUV的各种控制器提供了广泛的概述。从AUV实验的早期阶段，PID控制器就出现为一种简单且流行的选择 [33] - [36]。这些控制器被用于各种AUV子系统，包括速度控制、深度控制和舵机控制，通常采用线性PID设计 [37]、[38]、[39]。例如，REMUS AUV u

扰动和干扰是水下车辆的主要关注点。这会影晌系统的控制性能。一种常用于水下车辆控制以提供鲁棒性的方法是滑模控制（SMC）。该方法能够抵抗有界的外部扰动，并确保有限时间收敛。滑模控制（SMC）已被广泛应用于文献[44]-[47]中。在早期的应用中，SMC在一个非线性水下车辆模型上使用三种不同的二阶控制器对位置和速度进行测试，证明了其有效性[44]。在[48]中实现了一种改进的SMC，以应对潜水平面中变化的动力学和操作条件。[49]中的作者提出了一种带速度状态观测器的鲁棒自适应滑模控制律，用于3自由度水下自主车辆（AUV）模型，强调了其处理扰动和参数不确定性的能力。为一个6自由度模型设计了SMC控制器。

In [52], an adaptive SMC for pitch and yaw dynamics trajectory tracking is presented where the controller accounts for non-symmetric dead zones and unknown disturbances in the actuator. Additionally, an adaptive high-order sliding mode controller has been designed to track trajectories in the presence of persistent external disturbances [53]. Complex AUV scenarios, such as highly coupled nonlinearities, unknown system parameters, disturbance uncertainties, and input saturation are dealt with in [54], [55]. A method for handling the under-actuated AUV's trajectory tracking control with nonlinear model characteristics, model uncertainty, and roll motion effects is developed in [54]. Further, authors in [55], propose an adaptive backstepping sliding mode control method to mitigate uncertainties and disturbances, showcasing its superiority over conventional PID controllers in their comparative analysis. SMC has emerged as a versatile and robust control strategy for AUVs, enabling precise trajectory tracking and improved performance in challenging and uncertain underwater environments. Ongoing research in SMC and its combination with other control methodologies promises to further enhance AUV control capabilities and broaden their applications in various marine missions.

The focus of this book chapter is to provide a comprehensive study of modelling and control strategies for AUVs navigating in challenging underwater environments in the presence of ocean currents, parameters like buoyancy, hydro dynamic coefficients and centre of gravity variation. The foundation is laid through an intricate mathematical model that systematically integrates the impact of ocean currents into the AUV's dynamic behaviour, establishing a fundamental comprehension of its response within such complex surroundings. Further, the critical requirements and objectives for maintaining the desired level-flight of vehicles are investigated. During the level flight vehicle maintains constant velocity and orientation. This essentially involves finding one such control input that maintains the vehicle's equilibrium. It is seen in the literature the level-flight condition is largely ignored. Here, a detailed analysis of the level of flight dynamics is provided. Subsequently, the dynamics in the horizontal plane, vertical plane and speed subsystems are derived. These subsystems aid in simplifying the control design process and facilitate stability analysis of the nonlinear system. Further, the simulation section provides insights into the performance of linear control strategies, conventional nonlinear controllers, and robust SMC. The nonlinear controllers are evaluated under varying conditions, including parameter variations, ocean currents, and centre of gravity fluctuations. Finally, an analysis of the robustness of two controllers: conventional and SMC is presented with the lawn mowing manoeuvre.

The book chapter is organised as follows: in Section 2, we lay the foundation by developing a comprehensive mathematical model that incorporates the influence of ocean currents on AUV dynamics. Subsequently, in Section 3, we delve into the critical aspects of level flight dynamics and the linearization of the nonlinear model presented in Section 2. In Section 4, we present the simulation results obtained through three distinct control approaches: linear control, nonlinear control using conventional controller, and SMC. Finally, Section 5 gives the concluding remarks.

2 AUV Modelling

2.1 Research Vehicle Profile

Getting the hydrodynamic coefficients that are required for the mathematical formulation of the vehicle analysis of the vehicle profile is essential. It includes hull, nose and tail section profiles, and locations of various points such as the centre of gravity, the centre of buoyancy and fin parameters. The hull profile of the vehicle we are referring to is based on Myring hull profile [36]. This profile is designed such that there is a minimal drag coefficient for the required length-to-diameter ratio. The sub-parameters required to define the profile are a (length of nose section), b (length of mid-hull

在[52]中, 提出了一种用于俯仰和偏航动力学轨迹跟踪的自适应SMC控制器, 该控制器考虑了执行器中的非对称死区和未知干扰。此外, 还设计了一种自适应高阶滑模控制器, 以在持续外部干扰存在的情况下跟踪轨迹[53]。[54], [55]中处理了复杂的AUV场景, 如高度耦合非线性、未知系统参数、干扰不确定性和输入饱和。[54]中开发了一种处理欠驱动AUV轨迹跟踪控制的方法, 该方法考虑了非线性模型特性、模型不确定性和横滚运动影响。此外, [55]的作者提出了一种自适应反步滑模控制方法, 以减轻不确定性和干扰, 并在比较分析中展示了其优于传统PID控制器的性能。SMC已成为AUVs的一种多功能且稳健的控制策略, 使精确控制成为可能

本书本章的重点是提供对AUV (自主水下航行器) 在复杂水下环境中航行的建模和控制策略的全面研究, 考虑海流、浮力、水动力系数和重心变化等参数。通过一个复杂的数学模型奠定基础, 该模型系统地整合了海流对AUV动态行为的影响, 从而建立对其在如此复杂环境中响应的基本理解。此外, 还研究了维持车辆所需水平飞行的关键要求和目标。在水平飞行过程中, 车辆保持恒定速度和姿态。这本质上涉及找到能够维持车辆平衡的控制输入。文献中可以看到, 水平飞行条件在很大程度上被忽略。在这里, 提供了对水平飞行动力学的详细分析。随后, 动态

本书章节的组织如下: 在第2节中, 我们通过建立一个综合的数学模型来奠定基础, 该模型包含了洋流对AUV动力学的影响。随后, 在第3节中, 我们深入探讨平飞动力学的关键方面以及第2节中提出的非线性模型的线性化。在第4节中, 我们展示通过三种不同控制方法获得的仿真结果: 线性控制、使用传统控制器的非线性控制以及SMC。最后, 第5节给出结论性说明。

2 AUV Modelling

2.1 Research Vehicle Profile

获取用于车辆剖面车辆分析数学公式所需的水动力系数是必不可少的。它包括船体、船首和船尾部分的剖面, 以及各种点的位置, 例如重心、浮心和鳍参数。我们所指的车辆的船体剖面基于Myring船体剖面[36]。该剖面设计旨在使所需的长径比下阻力系数最小。定义该剖面所需的子参数包括 a (船首段长度), b (中船体段长度)

section), c (length of the tail section), d (maximum diameter), 2θ (included angle at the tip of tail) and n (a parameter which when varied will give different body shapes). Assuming an origin at the vehicle nose the equation of radius of curvature is given by

$$r(\Xi) = \frac{1}{2}d \left[1 - \left(\frac{\Xi + a_{\text{offset}} - a}{a} \right)^2 \right]^{\frac{1}{n}}, \quad (1)$$

where, r is the radius of the hull measured normally from the centerline, Ξ is the axial position along the centerline and a_{offset} is the missing vehicle nose length. Similarly, the tail section radius is given by

$$r(\Xi) = \frac{1}{2}d - \left[\frac{3d}{2c^2} - \frac{\tan \theta}{c} \right] (\Xi - l)^2 + \left[\frac{d}{c^3} - \frac{\tan \theta}{c^2} \right] (\Xi - l_f)^3 \quad (2)$$

where l is the forward body length and l_f is given as

$$l_f = a + b - a_{\text{offset}}. \quad (3)$$

All these parameters are tabulated in Table 6 in the appendix section. The general vehicle-related parameters along with the moment of inertia (MoI) parameters are tabulated in Table 2. MoI is with respect to the origin at the center of buoyancy (CB) which is located at 0.611 meters from the vehicle's nose. The location of the center of gravity (CG) wrt CB. The fin parameters and hull parameters are the same as in [36].

Table 2: Model Parameters of Research Vehicle

Description		Unit	Value
Vehicle mass (m)		kg	30.5
Seawater Density (ρ)		kg/m^3	1030
Vehicle Buoyancy (B)		N	306
MOI about x-axis (I_{xx})		$kg.m^2$	0.177
MOI about z-axis (I_{zz})		$kg.m^2$	3.45
Center of Bouancy (CB)	x_{cg}	m	0
	y_{cg}	m	0
	z_{cg}	m	0.0192

2.2 Mathematical Model

Mathematical modelling plays a fundamental role in the development and analysis of autonomous underwater vehicles (AUVs). AUVs are unmanned vehicles designed to operate autonomously in a challenging underwater environment, performing various tasks such as exploration, mapping, surveillance, and environmental monitoring. To ensure their efficient and safe operation, it is crucial to understand and predict the behaviour of these vehicles in different scenarios. A set of mathematical equations that capture the physical properties, kinematics, dynamics, and interactions with the environment is essential to gain insights into the behaviour and performance of AUVs. Accurate modelling of AUV dynamics and kinematics is fundamental to understanding their behaviour and

部分), c (尾部长度), d (最大直径), 2θ (尾尖包含角)以及 n (一个参数, 改变该参数会得到不同的机身形状)。假设原点位于飞行器鼻部, 曲率半径的方程为

$$r(\Xi) = \frac{1}{2}d \left[1 - \left(\frac{\Xi + a_{\text{offset}} - a}{a} \right)^2 \right]^{\frac{1}{n}}, \quad (1)$$

其中, r 是从中心线垂直测量的船体半径, Ξ 是沿中心线的轴向位置, a_{offset} 是缺失的车辆前端长度。类似地, 尾部截面的半径由以下公式给出

$$r(\Xi) = \frac{1}{2}d - \left[\frac{3d}{2c^2} - \frac{\tan \theta}{c} \right] (\Xi - l)^2 + \left[\frac{d}{c^3} - \frac{\tan \theta}{c^2} \right] (\Xi - l_f)^3 \quad (2)$$

其中 l 是前体长度, l_f 的值为

$$l_f = a + b - a_{\text{offset}}. \quad (3)$$

所有这些参数在附录部分的表6中列出。一般与车辆相关的参数以及惯性矩 (MoI) 参数在表2中列出。MoI 是相对于浮力中心 (CB) 原点的, 浮力中心位于距车辆前端0.611米的位置。重心 (CG) 相对于浮力中心的位置。鳍参数和船体参数与文献 [36] 中相同。

表 2: 研究车辆的模型参数

Description		Unit	Value
Vehicle mass (m)		kg	30.5
Seawater Density (ρ)		kg/m^3	1030
Vehicle Buoyancy (B)		N	306
MOI about x-axis (I_{xx})		$kg.m^2$	0.177
MOI about z-axis (I_{zz})		$kg.m^2$	3.45
Center of Bouancy (CB)	x_{cg}	m	0
	y_{cg}	m	0
	z_{cg}	m	0.0192

2.2 Mathematical Model

数学建模在自主水下航行器 (AUV) 的开发和分析中起着基础性作用。AUV 是一种无人驾驶车辆, 旨在自主地在复杂的水下环境中运行, 执行诸如勘探、测绘、监视和环境监测等各种任务。为了确保它们的高效和安全运行, 理解并预测这些车辆在不同情境下的行为至关重要。一组能够描述物理特性、运动学、动力学以及与环境相互作用的数学方程对于深入了解 AUV 的行为和性能是必不可少的。对 AUV 动力学和运动学的准确建模是理解其行为的基础, 并

interactions with the underwater environment. In summary, integrating accurate modelling and effective control strategies is indispensable for enhancing AUVs' autonomy, efficiency, and safety in underwater missions.

The mathematical model of an AUV encompasses several important aspects. The dynamics of an AUV consist of two components: kinematics, which defines the vehicle's motion in terms of position, orientation, and velocity, and kinetics, which defines the 6- degrees of freedom (DoF) with the forces and torques acting on the AUV, accounting for hydrodynamic effects, buoyancy, thrust forces, and external disturbances [36] and [56]. This dynamic representation allows for the study of the vehicle's response to different environmental conditions and control inputs.

According to SNAME (Society of Naval Architects and Marine Engineers), AUV is modelled with respect to two coordinate systems: body-fixed reference frame B and earth-fixed reference frame G . These coordinate systems are shown in Fig 1.

Figure 1 cord.jpg here

Fig. 1: Body-fixed and Earth-fixed Coordinate Systems

The 6 DoF model of AUV has six independent coordinates expressed in a body-fixed frame that determines the motion of the vehicle. The first three coordinates represent linear velocities $\boldsymbol{\nu}_1$ while the rest three represent angular velocities $\boldsymbol{\nu}_2$.

$$\boldsymbol{\nu} = \begin{bmatrix} \boldsymbol{\nu}_1 \\ \boldsymbol{\nu}_2 \end{bmatrix} = [u \ v \ w \ p \ q \ r]^T \quad (4)$$

where u , v , and w are linear velocities in the surge, sway, and heave axes respectively and p , q and r are angular velocities in roll, pitch and yaw respectively. The other coordinate system coincides with the earth's North-East-Down (NED) directions. The earth-fixed or inertial frame is fixed to a point on the water's surface. The position $\boldsymbol{\eta}_1$ and orientation $\boldsymbol{\eta}_2$ of the vehicle defined with respect to the earth-fixed frame respectively are,

$$\boldsymbol{\eta} = \begin{bmatrix} \boldsymbol{\eta}_1 \\ \boldsymbol{\eta}_2 \end{bmatrix} = [x \ y \ z \ \phi \ \theta \ \psi]^T \quad (5)$$

Lastly, total forces $\boldsymbol{\tau}_1$ and moments $\boldsymbol{\tau}_2$ acting on the vehicle are expressed in the body-fixed frame as

$$\boldsymbol{\tau} = \begin{bmatrix} \boldsymbol{\tau}_1 \\ \boldsymbol{\tau}_2 \end{bmatrix} = [X \ Y \ Z \ K \ M \ N]^T \quad (6)$$

where X , Y and Z represents forces and K , M and N represents moments in B . The following coordinate transform relates the time derivatives of $\boldsymbol{\eta}$ to $\boldsymbol{\nu}$:

$$\begin{bmatrix} \dot{\boldsymbol{\eta}}_1(t) \\ \dot{\boldsymbol{\eta}}_2(t) \end{bmatrix}_G = \begin{bmatrix} \mathbf{T}_B^G(\boldsymbol{\eta}_2) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{A}_B^G(\boldsymbol{\eta}_2) \end{bmatrix} \begin{bmatrix} \dot{\boldsymbol{\nu}}_1(t) \\ \dot{\boldsymbol{\nu}}_2(t) \end{bmatrix}_B \quad (7)$$

where $\mathbf{0}_{3 \times 3}$ is a null matrix of dimension 3, $\mathbf{T}_B^G$ transforms the linear velocities

$$\mathbf{T}_B^G(\boldsymbol{\eta}_2) = \begin{bmatrix} \cos \psi \cos \theta - \sin \psi \cos \phi + \cos \psi \sin \theta \sin \phi & \sin \psi \sin \phi + \cos \psi \sin \theta \cos \phi \\ \sin \psi \cos \theta & \cos \psi \cos \phi + \sin \psi \sin \theta \sin \phi & -\cos \psi \sin \phi + \sin \psi \sin \theta \cos \phi \\ -\sin \theta & \cos \theta \sin \phi & \cos \theta \cos \phi \end{bmatrix} \quad (8)$$

与水下环境的互动。总之，整合准确的建模和有效的控制策略对于提升自动水下车辆在水下任务中的自主性、效率 and 安全性是不可或缺的。

AUV 的数学模型包含几个重要方面。AUV 的动力学由两个部分组成：运动学，它以位置、方向和速度来定义车辆的运动；动力学，它以作用在 AUV 上的力和力矩来定义六自由度（DoF），并考虑水动力效应、浮力、推进力和外部扰动 [36] 和 [56]。这种动力学表示允许研究车辆对不同环境条件和控制输入的响应。

根据美国船舶与海洋工程师学会（SNAME），AUV 是相对于两个坐标系建模的：机体固定参考系 B 和地球固定参考系 G 。这些坐标系如图 1 所示。

图 1 cord.jpg 在这里

图 1：机体固定和地球固定坐标系

AUV 的六自由度模型在固定于车体的坐标系中有六个独立坐标，用于确定车辆的运动。前三个坐标表示线速度 ν_1 ，其余三个表示角速度 ν_2 。

$$\nu = \begin{bmatrix} \nu_1 \\ \nu_2 \end{bmatrix} = [u \ v \ w \ p \ q \ r]^T \quad (4)$$

其中 u 、 v 和 w 分别为沿纵向、横向和垂向的线速度， p 、 q 和 r 分别为滚转、俯仰和偏航方向的角速度。另一坐标系与地球的东北地向（NED）方向重合。固定于地球或惯性参考系的坐标系固定在水面上的一个点。相对于地球固定参考系定义的车辆位置 η_1 和姿态 η_2 分别为，

$$\eta = \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} = [x \ y \ z \ \phi \ \theta \ \psi]^T \quad (5)$$

最后，作用在车辆上的总力 τ_1 和力矩 τ_2 表示在车体固定坐标系中为

$$\tau = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = [X \ Y \ Z \ K \ M \ N]^T \quad (6)$$

其中 X 、 Y 和 Z 表示力， K 、 M 和 N 表示 B 中的力矩。以下坐标变换将 η 的时间导数与 ν 关联起来：

$$\begin{bmatrix} \dot{\eta}_1(t) \\ \dot{\eta}_2(t) \end{bmatrix}_G = \begin{bmatrix} T_B^G(\eta_2) & 0_{3 \times 3} \\ 0_{3 \times 3} & A_B^G(\eta_2) \end{bmatrix} \begin{bmatrix} \dot{\nu}_1(t) \\ \dot{\nu}_2(t) \end{bmatrix}_B \quad (7)$$

其中 $0_{3 \times 3}$ 是一个维度为 3 的零矩阵， T_B^G 转换线速度

$$T_B^G(\eta_2) = \begin{bmatrix} \cos \psi \cos \theta - \sin \psi \cos \phi + \cos \psi \sin \theta \sin \phi & \sin \psi \sin \phi + \cos \psi \sin \theta \cos \phi \\ \sin \psi \cos \theta & \cos \psi \cos \phi + \sin \psi \sin \theta \sin \phi & -\cos \psi \sin \phi + \sin \psi \sin \theta \cos \phi \\ -\sin \theta & \cos \theta \sin \phi & \cos \theta \cos \phi \end{bmatrix} \quad (8)$$

and the transformation matrix $T_B^G(\eta_2)$ satisfies orthogonality property,

$$T_B^G(\eta_2)^{-1} = T_B^G(\eta_2)^T$$

and A_B^G transformation matrix between time derivative of Euler angles and angular velocities and given by,

$$A_B^G(\eta_2) = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \frac{\sin \phi}{\cos \theta} & \frac{\cos \phi}{\cos \theta} \end{bmatrix} \quad (9)$$

The equations of motion can be derived using the Newton–Euler or Lagrange equations. In order to derive the equations of motion, it is necessary to study the motion of rigid bodies, hydro-statics and hydrodynamics. The system can be represented in the form of 6 DOF equations defined in body-fixed coordinates: wherein the three force equations in u , v , w axes are:

$$\begin{aligned} m[\dot{u} - vr + wq - x_g(q^2 - r^2) + y_g(pq - r^2) + z_g(pr + \dot{q})] &= X \\ m[\dot{v} - wp + ur - y_g(p^2 + r^2) + z_g(qr - \dot{p}) + x_g(pq + \dot{r})] &= Y \\ m[\dot{w} - uq + vp - z_g(p^2 + q^2) + x_g(rp - \dot{q}) + y_g(rq + \dot{p})] &= Z \end{aligned} \quad (10)$$

Similarly, moment balance in three axes leads to the following moment equations:

$$\begin{aligned} I_{xx} \dot{p} + (I_{zz} - I_{yy})qr + m[y_g(\dot{w} - uq + vp) - z_g(\dot{v} - wp + ur)] &= K \\ I_{yy} \dot{q} + (I_{xx} - I_{zz})rp + m[z_g(\dot{u} - vr + wq) - x_g(\dot{w} - uq + vp)] &= M \\ I_{zz} \dot{r} + (I_{yy} - I_{xx})pq + m[x_g(\dot{v} - wp + ur) - y_g(\dot{u} - vr + wq)] &= N \end{aligned} \quad (11)$$

where m is the mass of the AUV, x_g, y_g, z_g forms the location of center of gravity (C_g), I_{xx} , I_{yy} , I_{zz} are moments of the inertia of AUV, and the external forces :

$$\begin{aligned} X &= X_{HS} + X_{u|u}|u| + X_{\dot{u}}\dot{u} + X_{wq}wq + X_{q|q}|q| + X_{vr}vr + X_{rr}rr + X_{prop} \\ Y &= Y_{HS} + Y_{v|v}|v| + Y_{r|r}|r| + Y_{\dot{v}}\dot{v} + Y_{\dot{r}}\dot{r} + Y_{ur}ur + Y_{wp}wp + Y_{pq}pq + Y_{uv}uv \\ &\quad + Y_{uu\delta_r}u^2(\delta_{r_1} + \delta_{r_2}) \\ Z &= Z_{HS} + Z_{w|w}|w| + Z_{q|q}|q| + Z_{\dot{w}}\dot{w} + Z_{\dot{q}}\dot{q} + Z_{uq}uq + Z_{vp}vp + Z_{rp}rp + Z_{uw}uw \\ &\quad + Z_{uu\delta_s}u^2(\delta_{s_1} + \delta_{s_2}) \end{aligned} \quad (12)$$

An external moments K , M , N are given by:

$$\begin{aligned} K &= K_{HS} + K_{p|p}|p| + K_{\dot{p}}\dot{p} + K_{prop} + K_{roll}\delta_{roll} \\ M &= M_{HS} + M_{w|w}|w| + M_{q|q}|q| + M_{\dot{w}}\dot{w} + M_{\dot{q}}\dot{q} + M_{uq}uq + M_{vp}vp + M_{rp}rp + M_{uw}uw \\ &\quad + M_{uu\delta_s}u^2(\delta_{s_1} + \delta_{s_2}) \\ N &= N_{HS} + N_{v|v}|v| + N_{r|r}|r| + N_{\dot{v}}\dot{v} + N_{\dot{r}}\dot{r} + N_{ur}ur + N_{wp}wp + N_{pq}pq + N_{uv}uv \\ &\quad + N_{uu\delta_r}u^2(\delta_{r_1} + \delta_{r_2}) \end{aligned} \quad (13)$$

All the coefficients in eq. (12) and eq. (13) are given in Table 7. The matrix representation of eq. (10) and eq. (11),

$$\begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \\ \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} m - X_{\dot{u}} & 0 & 0 & 0 & mz_g & -my_g \\ 0 & m - Y_{\dot{v}} & 0 & -mz_g & 0 & mx_g - Y_{\dot{r}} \\ 0 & 0 & m - Z_{\dot{w}} & my_g & -mx_g - Z_{\dot{q}} & 0 \\ 0 & -mz_g & my_g & I_{xx} - K_{\dot{p}} & 0 & 0 \\ mz_g & 0 & -mx_g - M_{\dot{w}} & 0 & I_{yy} - M_{\dot{q}} & 0 \\ -my_g & mx_g - N_{\dot{v}} & 0 & 0 & 0 & I_{zz} - N_{\dot{r}} \end{bmatrix}^{-1} \begin{bmatrix} X \\ Y \\ Z \\ K \\ M \\ N \end{bmatrix} \quad (14)$$

并且变换矩阵 $T_B^G(\eta_2)$ 满足正交性性质,

$$T_B^G(\eta_2)^{-1} = T_B^G(\eta_2)^T$$

以及 A_B^G 变换矩阵, 它位于欧拉角的时间导数与角速度之间, 给出如下,

$$A_B^G(\eta_2) = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \frac{\sin \phi}{\cos \theta} & \frac{\cos \phi}{\cos \theta} \end{bmatrix} \quad (9)$$

运动方程可以使用牛顿-欧拉方程或拉格朗日方程来推导。为了推导运动方程, 有必要研究刚体运动、流体静力学和流体动力学。该系统可以表示为在机体固定坐标系中定义的六自由度方程的形式: 其中在 u, v, w 轴上的三个力方程为:

$$\begin{aligned} m[\dot{u} - vr + wq - x_g(q^2 - r^2) + y_g(pq - r^2) + z_g(pr + \dot{q})] &= X \\ m[\dot{v} - wp + ur - y_g(p^2 + r^2) + z_g(qr - \dot{p}) + x_g(pq + \dot{r})] &= Y \\ m[\dot{w} - uq + vp - z_g(p^2 + q^2) + x_g(rp - \dot{q}) + y_g(rq + \dot{p})] &= Z \end{aligned} \quad (10)$$

类似地, 三个轴上的力矩平衡导致以下力矩方程:

$$\begin{aligned} I_{xx} \dot{p} + (I_{zz} - I_{yy})qr + m[y_g(\dot{w} - uq + vp) - z_g(\dot{v} - wp + ur)] &= K \\ I_{yy} \dot{q} + (I_{xx} - I_{zz})rp + m[z_g(\dot{u} - vr + wq) - x_g(\dot{w} - uq + vp)] &= M \\ I_{zz} \dot{r} + (I_{yy} - I_{xx})pq + m[x_g(\dot{v} - wp + ur) - y_g(\dot{u} - vr + wq)] &= N \end{aligned} \quad (11)$$

其中 m 是 AUV 的质量, x_g, y_g, z_g 表示重心的位置 (C_g), I_{xx}, I_{yy}, I_{zz} 是 AUV 的惯性矩, 以及外力:

$$\begin{aligned} X &= X_{HS} + X_{u|u}|u| + X_{\dot{u}}\dot{u} + X_{wq}wq + X_{q|q}|q| + X_{vr}vr + X_{rr}rr + X_{prop} \\ Y &= Y_{HS} + Y_{v|v}|v| + Y_{r|r}|r| + Y_{\dot{v}}\dot{v} + Y_{\dot{r}}\dot{r} + Y_{ur}ur + Y_{wp}wp + Y_{pq}pq + Y_{uv}uv \\ &\quad + Y_{uu\delta_r}u^2(\delta_{r_1} + \delta_{r_2}) \\ Z &= Z_{HS} + Z_{w|w}|w| + Z_{q|q}|q| + Z_{\dot{w}}\dot{w} + Z_{\dot{q}}\dot{q} + Z_{uq}uq + Z_{vp}vp + Z_{rp}rp + Z_{uw}uw \\ &\quad + Z_{uu\delta_s}u^2(\delta_{s_1} + \delta_{s_2}) \end{aligned} \quad (12)$$

外力矩 K, M, N 由下式给出:

$$\begin{aligned} K &= K_{HS} + K_{p|p}|p| + K_{\dot{p}}\dot{p} + K_{prop} + K_{roll}\delta_{roll} \\ M &= M_{HS} + M_{w|w}|w| + M_{q|q}|q| + M_{\dot{w}}\dot{w} + M_{\dot{q}}\dot{q} + M_{uq}uq + M_{vp}vp + M_{rp}rp + M_{uw}uw \\ &\quad + M_{uu\delta_s}u^2(\delta_{s_1} + \delta_{s_2}) \\ N &= N_{HS} + N_{v|v}|v| + N_{r|r}|r| + N_{\dot{v}}\dot{v} + N_{\dot{r}}\dot{r} + N_{ur}ur + N_{wp}wp + N_{pq}pq + N_{uv}uv \\ &\quad + N_{uu\delta_r}u^2(\delta_{r_1} + \delta_{r_2}) \end{aligned} \quad (13)$$

方程 (12) 和方程 (13) 中的所有系数列于表 7。方程 (10) 和方程 (11) 的矩阵表示为

$$\begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \\ \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} m - X_{\dot{u}} & 0 & 0 & 0 & mz_g & -my_g \\ 0 & m - Y_{\dot{v}} & 0 & -mz_g & 0 & mx_g - Y_{\dot{r}} \\ 0 & 0 & m - Z_{\dot{w}} & my_g & -mx_g - Z_{\dot{q}} & 0 \\ 0 & -mz_g & my_g & I_{xx} - K_{\dot{p}} & 0 & 0 \\ mz_g & 0 & -mx_g - M_{\dot{w}} & 0 & I_{yy} - M_{\dot{q}} & 0 \\ -my_g & mx_g - N_{\dot{v}} & 0 & 0 & 0 & I_{zz} - N_{\dot{r}} \end{bmatrix}^{-1} \begin{bmatrix} X \\ Y \\ Z \\ K \\ M \\ N \end{bmatrix} \quad (14)$$

While eq. (14) offers a concise representation of the AUV's nonlinear model, its applicability may be limited when accounting for factors like ocean currents or external disturbances. The utilization of a modular system representation, on the other hand, presents distinct advantages during controller design, enabling the development of strategies specifically tailored to counteract the influence of external perturbations. In general, eq. (14) could be expressed in the following standard form,

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau} \quad (15)$$

where $\mathbf{M}$ is the mass and inertial matrix, $\mathbf{C}(\boldsymbol{\nu})$ is Coriolis and centripetal matrix, $\mathbf{D}(\boldsymbol{\nu})$ is damping matrix, $\mathbf{g}(\boldsymbol{\eta})$ is the hydrostatic vector and $\boldsymbol{\tau}$ is the control input vector. The details of the individual terms in eq. (15) are given below;

1. **Matrix $\mathbf{M}$:** It is composed of rigid-body system inertia matrix $\mathbf{M}_{RB}$ and added mass matrix $\mathbf{M}_A$,

$$\mathbf{M} = \mathbf{M}_{RB} + \mathbf{M}_A \quad (16)$$

The rigid-body system inertia matrix $\mathbf{M}_{RB}$

$$\mathbf{M}_{RB} = \begin{bmatrix} m & 0 & 0 & 0 & mz_g & -my_g \\ 0 & m & 0 & -mz_g & 0 & mx_g \\ 0 & 0 & m & my_g & -mx_g & 0 \\ 0 & -mz_g & my_g & I_{xx} & 0 & 0 \\ mz_g & 0 & -mx_g & 0 & I_{yy} & 0 \\ -my_g & mx_g & 0 & 0 & 0 & I_{zz} \end{bmatrix} \quad (17)$$

The amount of water that is added to the vehicle's mass as it accelerates is known as added mass. When a vehicle accelerates or decelerates, the volume of the surrounding fluid must be changed, which results in a pressure-induced force and moment on the body. This effect is substantial for objects moving through the water, and its contribution can be described as an apparent rise in the real mass of the submerged object. The added mass matrix $\mathbf{M}_A$ expressed as,

$$\mathbf{M}_A = - \begin{bmatrix} X_{\ddot{u}} & 0 & 0 & 0 & 0 & 0 \\ 0 & Y_{\ddot{v}} & 0 & 0 & 0 & Y_{\ddot{r}} \\ 0 & 0 & Z_{\ddot{w}} & 0 & Z_{\ddot{q}} & 0 \\ 0 & 0 & 0 & K_{\ddot{p}} & 0 & 0 \\ 0 & 0 & M_{\ddot{w}} & 0 & M_{\ddot{q}} & 0 \\ 0 & N_{\ddot{v}} & 0 & 0 & 0 & N_{\ddot{r}} \end{bmatrix} \quad (18)$$

where $X_{\ddot{u}}$, $Y_{\ddot{v}}$, $Y_{\ddot{r}}$, $Z_{\ddot{w}}$, $Z_{\ddot{q}}$, $K_{\ddot{p}}$, $M_{\ddot{w}}$, $M_{\ddot{q}}$, $N_{\ddot{v}}$, $N_{\ddot{r}}$ are hydrodynamic added mass coefficients. Calculation of these coefficients for a generic shape is a challenging problem that requires the use of CFD techniques and experimental verification. However, there are a number of streamlined techniques that offer suitable first-order approximations for frequently seen shapes (such as spheres, cylinders, plates, etc.) [36].

2. **Coriolis and Centripetal matrix $\mathbf{C}(\boldsymbol{\nu})$:** This matrix is composed of two entities, the rigid body Coriolis and centripetal matrix $\mathbf{C}_{RB}(\boldsymbol{\nu})$ and Coriolis and centripetal effect due to added mass $\mathbf{C}_A(\boldsymbol{\nu})$.

$$\mathbf{C}(\boldsymbol{\nu}) = \mathbf{C}_{RB}(\boldsymbol{\nu}) + \mathbf{C}_A(\boldsymbol{\nu}) \quad (19)$$

The matrix captures the effects of the Coriolis and centripetal force on the motion of the AUV. The Coriolis force arises due to the vehicle's motion in a rotating frame of reference, such as the

虽然公式(14)提供了AUV非线性模型的简明表示，但在考虑洋流或外部扰动等因素时，其适用性可能有限。另一方面，采用模块化系统表示在控制器设计过程中具有明显优势，使得开发专门针对抵消外部扰动影响的策略成为可能。通常，公式(14)可以用以下标准形式表示，

$$M\dot{\nu} + C(\nu)\nu + D(\nu)\nu + g(\eta) = \tau \quad (15)$$

其中 M 是质量和惯性矩阵， $C(\nu)$ 是科里奥利和向心矩阵， $D(\nu)$ 是阻尼矩阵， $g(\eta)$ 是静水矢量， τ 是控制输入向量。方程 (15) 中各个项的详细内容如下；

1. **Matrix M :** 它由刚体系统惯性矩阵 M_{RB} 和附加质量矩阵 M_A 组成，

$$M = M_{RB} + M_A \quad (16)$$

刚体系统惯性矩阵 M_{RB}

$$M_{RB} = \begin{bmatrix} m & 0 & 0 & 0 & mz_g & -my_g \\ 0 & m & 0 & -mz_g & 0 & mx_g \\ 0 & 0 & m & my_g & -mx_g & 0 \\ 0 & -mz_g & my_g & I_{xx} & 0 & 0 \\ mz_g & 0 & -mx_g & 0 & I_{yy} & 0 \\ -my_g & mx_g & 0 & 0 & 0 & I_{zz} \end{bmatrix} \quad (17)$$

随着车辆加速而加入车辆质量的水量被称为附加质量。当车辆加速或减速时，周围流体的体积必须发生变化，这会在物体上产生由压力引起的力和力矩。这种效应对于在水中运动的物体来说是显著的，其贡献可以描述为潜水物体真实质量的表现增加。附加质量矩阵 M_A 表示为，

$$M_A = - \begin{bmatrix} X_{\dot{u}} & 0 & 0 & 0 & 0 & 0 \\ 0 & Y_{\dot{v}} & 0 & 0 & 0 & Y_{\dot{r}} \\ 0 & 0 & Z_{\dot{w}} & 0 & Z_{\dot{q}} & 0 \\ 0 & 0 & 0 & K_{\dot{p}} & 0 & 0 \\ 0 & 0 & M_{\dot{w}} & 0 & M_{\dot{q}} & 0 \\ 0 & N_{\dot{v}} & 0 & 0 & 0 & N_{\dot{r}} \end{bmatrix} \quad (18)$$

其中 $X_{\dot{u}}$, $Y_{\dot{v}}$, $Y_{\dot{r}}$, $Z_{\dot{w}}$, $Z_{\dot{q}}$, $K_{\dot{p}}$, $M_{\dot{w}}$, $M_{\dot{q}}$, $N_{\dot{v}}$, $N_{\dot{r}}$ 是流体动力附加质量系数。对于通用形状，这些系数的计算是一个具有挑战性的问题，需要使用计算流体力学 (CFD) 技术和实验验证。然而，有许多简化的技术可以为常见形状（如球体、圆柱体、板等）提供合适的一阶近似 [36]

。

2. **Coriolis and Centripetal matrix $C(\nu)$:** 该矩阵由两个部分组成，一个是刚体科里奥利和向心矩阵 $C_{RB}(\nu)$ ，另一个是由于附加质量引起的科里奥利和向心效应 $C_A(\nu)$ 。

$$C(\nu) = C_{RB}(\nu) + C_A(\nu) \quad (19)$$

该矩阵捕捉了科里奥利力和向心力对自主水下航行器运动的影响。科里奥利力是由于车辆在旋转参考系中运动而产生的，例如

Earth. While centripetal forces arise due to the vehicle's curved motion or manoeuvres. As the AUV moves through the water, its motion experiences a force that tends to deflect its trajectory sideways. In general, the matrix C_{RB} in eq. (19) is given by,

$$C_{RB}(\nu) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ -m(y_g q + z_g r) & m(y_g p + w) & m(z_g p - v) \\ m(x_g q - w) & -m(z_g r + x_g p) & m(z_g q + u) \\ m(x_g r + v) & m(y_g r - u) & -m(x_g p + y_g q) \end{bmatrix} \quad (20)$$

$$\begin{bmatrix} m(y_g q + z_g r) & -m(x_g q - w) & -m(x_g r + v) \\ -m(y_g p + w) & m(z_g r + x_g p) & -m(y_g r - u) \\ -m(z_g p - v) & -m(z_g q + u) & m(x_g p + y_g q) \\ 0 & -I_{yz}q - I_{xz}p + I_{zz}r & I_{yz}r + I_{xy}p - I_{yy}q \\ I_{yz}q + I_{xz}p - I_{zz}r & 0 & -I_{xz}r - I_{xy}q + I_{xx}p \\ -I_{yz}r - I_{xy}p + I_{yy}q & I_{xz}r + I_{xy}q - I_{xx}p & 0 \end{bmatrix}$$

Due to the fact that the vehicle has two axial planes of symmetry and homogeneous mass distribution, I_{xy} , I_{xz} and I_{yz} are negligible compared to I_{xx} , I_{yy} and I_{zz} . They are assumed to be zeros. Assuming that the B -frame coordinate origin is set in the centre line of the AUV, hence, $y_g = 0$. The matrix C_A in eq. (19) is analogous to M_A in eq. (18) which takes in the additional effects due to the motion of the vehicle in the fluid.

$$C_A(\nu) = \begin{bmatrix} 0 & 0 & 0 & 0 & -Z_{\dot{w}}w & Y_{\dot{v}}v \\ 0 & 0 & 0 & Z_{\dot{w}}w & 0 & -X_{\dot{u}}u \\ 0 & 0 & 0 & -Y_{\dot{v}}v & X_{\dot{u}}u & 0 \\ 0 & -Z_{\dot{w}}w & Y_{\dot{v}}v & 0 & -N_{\dot{r}}r & M_{\dot{q}}q \\ Z_{\dot{w}}w & 0 & -X_{\dot{u}}u & N_{\dot{r}}r & 0 & -K_{\dot{p}}p \\ -Y_{\dot{v}}v & X_{\dot{u}}u & 0 & -M_{\dot{q}}q & K_{\dot{p}}p & 0 \end{bmatrix} \quad (21)$$

3. **Hydrodynamic Damping matrix $D(\nu)$:** Hydrodynamic damping is a critical aspect of the hydrodynamic forces acting on an AUV. It is usually written as the sum of linear and non-linear damping,

$$D(\nu) = D_L + D_n(\nu) \quad (22)$$

The linear damping matrix D_L is due to potential damping and skin friction and the nonlinear damping matrix $D_n(\nu)$ is due to quadratic damping and higher-order terms. As mentioned it arises due to various factors, including skin friction, vortex shedding, and lifting forces. Skin friction is the resistance to motion experienced by the AUV as it interacts with the water. When the AUV's hull moves through the water, the friction between the hull's surface and the water generates drag forces. Skin friction damping is influenced by factors such as the hull's shape, the roughness of the surface, and the vehicle's speed. An AUV with a propeller moves through the water, it can generate vortices in its wake due to the flow separation around the vehicle's body. These vortices can lead to vortex shedding, a phenomenon where vortices are periodically shed from the AUV's surface. Vortex shedding introduces additional drag forces on the AUV. Lifting forces, also known as hydrodynamic lift, occur when there is a pressure difference between the upper and lower surfaces of the AUV's hull. The linear damping matrix D_L in eq. (22) can be

地球。当车辆发生曲线运动或操作时，会产生向心力。当AUV在水中运动时，其运动会受到一种力的影响，这种力倾向于使其轨迹向侧面偏转。通常情况下，方程(19)中的矩阵 C_{RB} 表示为，

$$C_{RB}(\nu) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ -m(y_g q + z_g r) & m(y_g p + w) & m(z_g p - v) \\ m(x_g q - w) & -m(z_g r + x_g p) & m(z_g q + u) \\ m(x_g r + v) & m(y_g r - u) & -m(x_g p + y_g q) \end{bmatrix} \quad (20)$$

$$\begin{bmatrix} m(y_g q + z_g r) & -m(x_g q - w) & -m(x_g r + v) \\ -m(y_g p + w) & m(z_g r + x_g p) & -m(y_g r - u) \\ -m(z_g p - v) & -m(z_g q + u) & m(x_g p + y_g q) \\ 0 & -I_{yz}q - I_{xz}p + I_{zz}r & I_{yz}r + I_{xy}p - I_{yy}q \\ I_{yz}q + I_{xz}p - I_{zz}r & 0 & -I_{xz}r - I_{xy}q + I_{xx}p \\ -I_{yz}r - I_{xy}p + I_{yy}q & I_{xz}r + I_{xy}q - I_{xx}p & 0 \end{bmatrix}$$

由于该车辆具有两个轴对称平面且质量分布均匀， I_{xy} 、 I_{xz} 和 I_{yz} 相较于 I_{xx} 、 I_{yy} 和 I_{zz} 可忽略不计。假设它们为零。假设 B 坐标系原点设置在AUV的中心线上，因此 $y_g = 0$ 。方程(19)中的矩阵 C_A 类似于方程(18)中的 M_A ，它考虑了车辆在流体中运动的附加影响。

$$C_A(\nu) = \begin{bmatrix} 0 & 0 & 0 & 0 & -Z_{\dot{w}}w & Y_{\dot{v}}v \\ 0 & 0 & 0 & Z_{\dot{w}}w & 0 & -X_{\dot{u}}u \\ 0 & 0 & 0 & -Y_{\dot{v}}v & X_{\dot{u}}u & 0 \\ 0 & -Z_{\dot{w}}w & Y_{\dot{v}}v & 0 & -N_{\dot{r}}r & M_{\dot{q}}q \\ Z_{\dot{w}}w & 0 & -X_{\dot{u}}u & N_{\dot{r}}r & 0 & -K_{\dot{p}}p \\ -Y_{\dot{v}}v & X_{\dot{u}}u & 0 & -M_{\dot{q}}q & K_{\dot{p}}p & 0 \end{bmatrix} \quad (21)$$

3. Hydrodynamic Damping matrix $D(\nu)$: 水动力阻尼是作用于自主水下航行器（AUV）的水动力力的一个关键方面。它通常表示为线性阻尼和非线性阻尼的总和，

$$D(\nu) = D_L + D_n(\nu) \quad (22)$$

线性阻尼矩阵 D_L 是由潜在阻尼和表面摩擦引起的，而非线性阻尼矩阵 $D_n(\nu)$ 是由二次阻尼和高阶项引起的。如前所述，它是由各种因素产生的，包括表面摩擦、涡旋脱落和升力。表面摩擦是自主水下航行器（AUV）与水接触时所经历的运动阻力。当AUV的船体在水中移动时，船体表面与水之间的摩擦会产生阻力。表面摩擦阻尼受船体形状、表面粗糙度和车辆速度等因素的影响。当带螺旋桨的AUV在水中移动时，由于车辆周围的流动分离，船体后方可能会产生涡旋。这些涡旋可能导致涡旋脱落现象，即涡旋周期性地从AUV表面脱落。涡旋脱落会对AUV产生额外的阻力。升力，也称为水动力升力，发生在

written as,

$$\mathbf{D}_L = - \begin{bmatrix} X_u & 0 & 0 & 0 & 0 & 0 \\ 0 & Y_v & 0 & Y_p & 0 & Y_r \\ 0 & 0 & Z_w & 0 & Z_q & 0 \\ 0 & K_v & 0 & K_p & 0 & K_r \\ 0 & 0 & M_w & 0 & M_q & 0 \\ 0 & N_v & 0 & N_p & 0 & N_r \end{bmatrix}$$

Certain cross-terms can be neglected when AUV is operating near the equilibrium point. The nonlinear damping matrix $\mathbf{D}_n(\boldsymbol{\nu})$ has two components,

$$\mathbf{D}_n(\boldsymbol{\nu}) = \mathbf{D}_{n_1}(\boldsymbol{\nu}) + \mathbf{D}_{n_2}(\boldsymbol{\nu}) \quad (23)$$

where, $\mathbf{D}_{n_1}(\boldsymbol{\nu})$ is the expression of crossflow drag terms,

$$\mathbf{D}_{n_1}(\boldsymbol{\nu}) = - \begin{bmatrix} X_{|u|u}|u| & 0 & 0 & 0 & 0 & 0 \\ 0 & Y_{|v|v}|v| & 0 & 0 & 0 & Y_{|r|r}|r| \\ 0 & 0 & Z_{|w|w}|w| & 0 & Z_{|q|q}|q| & 0 \\ 0 & 0 & 0 & K_{|p|p}|p| & 0 & 0 \\ 0 & 0 & M_{|w|w}|w| & 0 & M_{|q|q}|q| & 0 \\ 0 & N_{|v|v}|v| & 0 & 0 & 0 & N_{|r|r}|r| \end{bmatrix} \quad (24)$$

and $\mathbf{D}_{n_2}(\boldsymbol{\nu})$ deals with the lift force and moments on the hull and fins of AUV,

$$\mathbf{D}_{n_2}(\boldsymbol{\nu}) = - \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & Y_{uv}u & 0 & 0 & 0 & Y_{ur}u \\ 0 & 0 & Z_{uw}u & 0 & Z_{uq}u & 0 \\ 0 & 0 & 0 & K_{up}u & 0 & 0 \\ 0 & 0 & M_{uw}u & 0 & M_{uq}u & 0 \\ 0 & N_{uv}u & 0 & 0 & 0 & N_{ur}u \end{bmatrix} \quad (25)$$

4. **Hydrostatic/ Resorting force vector $\mathbf{g}(\boldsymbol{\eta})$:** The vehicle experiences hydrostatic forces and moments as a result of the combined effects of the vehicle weight and buoyancy. The vehicle weight is given by $W = mg$ and buoyancy is expressed as $B = \rho V_m g$, where ρ is the density of the surrounding fluid and V_m is the total volume displaced by the vehicle. These forces and moments are expressed in B . The gravitational and buoyancy forces acting on the centre of gravity ($\mathbf{c}_g$) and centre of buoyancy ($\mathbf{c}_b$) of an AUV are shown in Fig.2

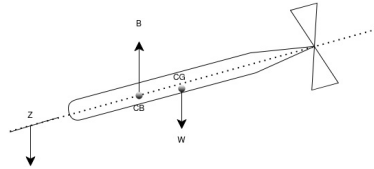


Fig. 2: Hydrostatic forces acting on AUV [57]

写作,

$$D_L = - \begin{bmatrix} X_u & 0 & 0 & 0 & 0 & 0 \\ 0 & Y_v & 0 & Y_p & 0 & Y_r \\ 0 & 0 & Z_w & 0 & Z_q & 0 \\ 0 & K_v & 0 & K_p & 0 & K_r \\ 0 & 0 & M_w & 0 & M_q & 0 \\ 0 & N_v & 0 & N_p & 0 & N_r \end{bmatrix}$$

当AUV在接近平衡点运行时, 可以忽略某些交叉项。非线性阻尼矩阵 $D_n(\nu)$ 有两个组成部分,

$$D_n(\nu) = D_{n_1}(\nu) + D_{n_2}(\nu) \quad (23)$$

其中, $D_{n_1}(\nu)$ 是横向流阻力项的表达式,

$$D_{n_1}(\nu) = - \begin{bmatrix} X_{|u|u}|u| & 0 & 0 & 0 & 0 & 0 \\ 0 & Y_{|v|v}|v| & 0 & 0 & 0 & Y_{|r|r}|r| \\ 0 & 0 & Z_{|w|w}|w| & 0 & Z_{|q|q}|q| & 0 \\ 0 & 0 & 0 & K_{|p|p}|p| & 0 & 0 \\ 0 & 0 & M_{|w|w}|w| & 0 & M_{|q|q}|q| & 0 \\ 0 & N_{|v|v}|v| & 0 & 0 & 0 & N_{|r|r}|r| \end{bmatrix} \quad (24)$$

以及 $D_{n_2}(\nu)$ 处理 AUV 船体和鳍上的升力和力矩,

$$D_{n_2}(\nu) = - \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & Y_{uv}u & 0 & 0 & 0 & Y_{ur}u \\ 0 & 0 & Z_{uw}u & 0 & Z_{uq}u & 0 \\ 0 & 0 & 0 & K_{up}u & 0 & 0 \\ 0 & 0 & M_{uw}u & 0 & M_{uq}u & 0 \\ 0 & N_{uv}u & 0 & 0 & 0 & N_{ur}u \end{bmatrix} \quad (25)$$

四Hydrostatic/ Resorting force vector $g(\eta)$: 由于车辆重量和浮力的综合作用, 车辆会受到静水力的力和力矩的作用。车辆重量由 $W = mg$ 给出, 浮力表示为 $B = \rho V_m g$, 其中 ρ 是周围流体的密度, V_m 是车辆排开的总体积。这些力和力矩以 B 表示。作用在 AUV 重心 (c_g) 和浮心 (c_b) 的重力和浮力如图 2 所示

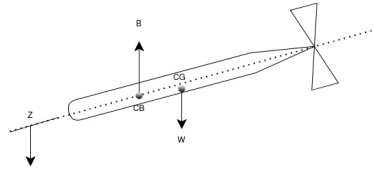


图2: 作用在AUV上的静水力 [57]

$$g(\eta) = \begin{bmatrix} X_{HS} \\ Y_{HS} \\ Z_{HS} \\ K_{HS} \\ M_{HS} \\ N_{HS} \end{bmatrix} = \begin{bmatrix} -(W-B)\sin\theta \\ (W-B)\cos\theta\sin\phi \\ (W-B)\cos\theta\cos\phi \\ -(y_gW - y_bB)\cos\theta\cos\phi - (z_gW - z_bB)\cos\theta\sin\phi \\ -(z_gW - z_bB)\sin\theta - (x_gW - x_bB)\cos\theta\cos\phi \\ -(x_gW - x_bB)\cos\theta\sin\phi - (y_gW - y_bB)\sin\theta \end{bmatrix} \quad (26)$$

where, $\mathbf{c}_b = [x_b \ y_b \ z_b]^T$ are vehicle's centers of buoyancy.

5. **Control Input τ :** It consists of the terms related to the thrust and torque of the propeller and control surfaces: sterns δ_{s1} and δ_{s2} and rudders δ_{r1} and δ_{r2} .

$$\tau = \begin{bmatrix} X_{prop} \\ Y_r \\ Z_s \\ K_{prop} \\ M_s \\ N_r \end{bmatrix} = \begin{bmatrix} (1 - \tau_p) T_{n|n|} n |n| \\ Y_{uu\delta_r} u^2 (\delta_{r1} + \delta_{r2}) \\ Z_{uu\delta_s} u^2 (\delta_{s1} + \delta_{s2}) \\ Q_{n|n|} n |n| + K_{roll} \delta_{roll} \\ M_{uu\delta_s} u^2 (\delta_{s1} + \delta_{s2}) \\ N_{uu\delta_r} u^2 (\delta_{r1} + \delta_{r2}) \end{bmatrix} \quad (27)$$

where $Y_{uu\delta_r}, Z_{uu\delta_s}, m_{uu\delta_s}, N_{uu\delta_r}$ are fin lift coefficients and X_{prop} and K_{prop} are the thrust and the torque of the propeller system. The thrust and torque are obtained from the model basin or thrust bench. Axial inflow u_p is the main reason for thrust losses. This is generally neglected while designing the speed controller, eventually leading to thrust degradation. To account for this [58] presented a model that justified the anomalies between experimental and mathematical thrust data. The two-state model is:

$$\begin{aligned} J_m \dot{n} &= Q - K_n n - Q_{n|n|} n |n| \\ m_f \dot{u}_p &= T_{n|n|} n |n| - d_f u_p - d_f |u_p| (u_p - (1 - w_p)u) \end{aligned} \quad (28)$$

where d_{f0} and d_f are damping coefficients given by,

$$\begin{aligned} d_{f0} &= \frac{-2X_{u|u|}}{(1 - \tau_p)(1 + a_p)(1 - w_p)}, \\ d_f &= \frac{-X_{u|u|}}{(1 - \tau_p)(1 + a_p)(1 - w_p)^2}, \end{aligned} \quad (29)$$

where Q is propeller torque, J_m is propeller inertia, m_f is mass in control volume, $T_{n|n|}$ and $Q_{n|n|}$ are thrust and torque coefficients respectively, τ_p is thrust reduction factor, a_p is axial flow parameter, w_p is thrust wake fraction number. Another simple model for the propulsion system, treats the propeller as a source of constant thrust and torque. The experimental coefficient data obtained by [36] are used for mathematical simulation. Based on the experimental data at different speeds, the thrust and torque can be calculated using a second-order polynomial for the given speed of the propeller of the AUV.

Propeller force on AUV in X-direction as a function of rotational speed can be approximated as,

$$X_{prop} = a_1 * n^2 + a_2 * n + a_3 \quad (30)$$

and similarly, the torque along the X-direction as a result of the propeller rotation be,

$$K_{prop} = b_1 * n^2 + b_2 * n + b_3 \quad (31)$$

$$g(\eta) = \begin{bmatrix} X_{HS} \\ Y_{HS} \\ Z_{HS} \\ K_{HS} \\ M_{HS} \\ N_{HS} \end{bmatrix} = \begin{bmatrix} -(W-B)\sin\theta \\ (W-B)\cos\theta\sin\phi \\ (W-B)\cos\theta\cos\phi \\ -(y_g W - y_b B)\cos\theta\cos\phi - (z_g W - z_b B)\cos\theta\sin\phi \\ -(z_g W - z_b B)\sin\theta - (x_g W - x_b B)\cos\theta\cos\phi \\ -(x_g W - x_b B)\cos\theta\sin\phi - (y_g W - y_b B)\sin\theta \end{bmatrix} \quad (26)$$

其中, $\mathbf{c}_b = [x_b \ y_b \ z_b]^T$ 是船舶的浮力中心。

5. Control Input τ : 它包括与螺旋桨和控制面的推力与扭矩相关的术语: 船尾 δ_{s_1} 和 δ_{s_2} 以及舵 δ_{r_1} 和 δ_{r_2} 。

$$\tau = \begin{bmatrix} X_{prop} \\ Y_r \\ Z_s \\ K_{prop} \\ M_s \\ N_r \end{bmatrix} = \begin{bmatrix} (1 - \tau_p) T_{n|n|} n |n| \\ Y_{uu\delta_r} u^2 (\delta_{r_1} + \delta_{r_2}) \\ Z_{uu\delta_s} u^2 (\delta_{s_1} + \delta_{s_2}) \\ Q_{n|n|} n |n| + K_{roll} \delta_{roll} \\ M_{uu\delta_s} u^2 (\delta_{s_1} + \delta_{s_2}) \\ N_{uu\delta_r} u^2 (\delta_{r_1} + \delta_{r_2}) \end{bmatrix} \quad (27)$$

其中 $Y_{uu\delta_r}, Z_{uu\delta_s}, m_{uu\delta_s}, N_{uu\delta_r}$ 是鳍升力系数, X_{prop} 和 K_{prop} 是推进器系统的推力和扭矩。推力和扭矩来自模型池或推力测试台。轴向流入 u_p 是推力损失的主要原因。在设计速度控制器时通常忽略这一点, 最终导致推力下降。为了解释这一点, [58] 提出了一个模型, 用以说明实验推力数据与数学推力数据之间的异常。该两状态模型为:

$$\begin{aligned} J_m \dot{n} &= Q - K_n n - Q_{n|n|} n |n| \\ m_f \dot{u}_p &= T_{n|n|} n |n| - d_f u_p - d_f |u_p| (u_p - (1 - w_p) u) \end{aligned} \quad (28)$$

其中 d_{f0} 和 d_f 是阻尼系数, 由以下公式给出,

$$\begin{aligned} d_{f0} &= \frac{-2X_{u|u|}}{(1 - \tau_p)(1 + a_p)(1 - w_p)}, \\ d_f &= \frac{-X_{u|u|}}{(1 - \tau_p)(1 + a_p)(1 - w_p)^2}, \end{aligned} \quad (29)$$

Q 是螺旋桨扭矩, J_m 是螺旋桨转动惯量, m_f 是控制体积内的质量, $T_{n|n|}$ 和 $Q_{n|n|}$ 分别是推力和扭矩系数, τ_p 是推力减小因子, a_p 是轴向流动参数, w_p 是推力尾迹系数。另一个简单的推进系统模型, 将螺旋桨视为恒定推力和扭矩的来源。数学模拟使用了 [36] 获得的实验系数数据。基于不同速度下的实验数据, 可以使用二次多项式计算AUV螺旋桨在给定速度下的推力和扭矩。

AUV在X方向上的螺旋桨推力作为转速的函数可以近似表示为,

$$X_{prop} = a_1 * n^2 + a_2 * n + a_3 \quad (30)$$

同样, 由螺旋桨旋转产生的沿X方向的扭矩为,

$$K_{prop} = b_1 * n^2 + b_2 * n + b_3 \quad (31)$$

where n is the speed of the propeller in rotations per second (RPS) and (a_1, a_2, a_3) and (b_1, b_2, b_3) are constants. To simplify the equations we use second-order polynomial approximation as the second-order curve fitting for thrust vs. speed gives an accuracy of 98.32%.

Ocean currents: The nonlinear state-space model in eq. (15) has no effect on external disturbances. However, in practical scenarios, the system may be subjected to temporary or persistent ocean currents. Ocean currents are driven by gravity, wind (Coriolis Effect) and water density. They can significantly affect the speed, trajectory, and energy consumption. They play a pivotal role in underwater vehicles' dynamics and navigation. Incorporating ocean current models enables us to evaluate the intricate interplay between the vehicle's intended path and the prevailing oceanic flows. We modified the mathematical model eq. (15) to include the effect of ocean currents and given by,

$$M\dot{\boldsymbol{\nu}}_r + \mathbf{C}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau} \quad (32)$$

where, the relative velocity $\boldsymbol{\nu}_r = \boldsymbol{\nu} - \boldsymbol{\nu}_c = [u - u_c \ v - v_c \ w - w_c \ p \ q \ r]^T = [u_r \ v_r \ w_r \ p \ q \ r]^T$ and the ocean component u_c, v_c, w_c is given by,

$$\begin{aligned} u_c &= U_c \cos \alpha_c \cos \beta_c \\ v_c &= U_c \sin \beta_c \\ w_c &= U_c \sin \alpha_c \cos \beta_c \end{aligned} \quad (33)$$

Here, U_c ocean current speed with its direction relative to the AUV is expressed in two angles: angle of attack α_c and side slip angle β_c . The model in eq. (33) is a 3-D ocean current model. The ocean current velocity is usually modelled as a first-order system with Gaussian white noise input η [56]:

$$\dot{U}_c + \zeta U_c = \mu \quad (34)$$

We choose $\zeta \geq 0$ such that the eq. (34) results in a stable system. Apart from U_c , we can associate dynamics to α_c and β_c .

3 Level Flight Dynamics and Linearization

3.1 Level flight dynamics

Level flight is a condition where a constant set of inputs is given to the vehicle without any disturbances and the vehicle will maintain fixed velocity and orientation. It is also known as *trim* condition. It is a crucial process that involves adjusting the vehicle's configuration and control settings to achieve a state of equilibrium and stability in the water. An AUV when it is below the water surface has many forces acting on it simultaneously. In order for an AUV to satisfy the level flight conditions as mentioned earlier, all these forces and moments need to be balanced [57], which will further result in zero acceleration and zero rates of rotation about all three axes. Mathematically this becomes a problem of estimating the control parameters which will result in this state. The control surfaces here are, stern angles, rudder angles and propeller

$$\mathbf{u}_i = [n \ \delta_{s_1} \ \delta_{s_2} \ \delta_{r_1} \ \delta_{r_2}]^T$$

The above vector consists of propeller RPM, two stern angles and two rudder angles. In order to solve for trim, the implementation of a mathematical model of the vehicle is done in the form of a function taking vehicle states along with control inputs and giving the rate of change of all

其中 n 是螺旋桨的转速, 单位为每秒转数 (RPS), (a_1, a_2, a_3) 和 (b_1, b_2, b_3) 是常数。为简化方程, 我们使用二次多项式近似, 因为推力与速度的二次曲线拟合的精度为 98.32%。

Ocean currents: 方程 (15) 中的非线性状态空间模型对外部扰动没有影响。然而, 在实际情况下, 系统可能会受到暂时或持续的洋流影响。洋流由重力、风 (科里奥利力效应) 和水的密度驱动。它们可以显著影响速度、轨迹和能耗。在水下车辆的动力学和导航中, 洋流起着关键作用。引入洋流模型使我们能够评估车辆预定路径与现有海洋流之间的复杂相互作用。我们修改了数学模型方程 (15), 以包括洋流的影响, 如下所示,

$$M\dot{\boldsymbol{\nu}}_r + \mathbf{C}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau} \quad (32)$$

其中, 相对速度 $\boldsymbol{\nu}_r = \boldsymbol{\nu} - \boldsymbol{\nu}_c = [u - u_c \ v - v_c \ w - w_c \ p \ q \ r]^T = [u_r \ v_r \ w_r \ p \ q \ r]^T$ 和海洋成分 u_c, v_c, w_c 表示为,

$$\begin{aligned} u_c &= U_c \cos \alpha_c \cos \beta_c \\ v_c &= U_c \sin \beta_c \\ w_c &= U_c \sin \alpha_c \cos \beta_c \end{aligned} \quad (33)$$

这里, U_c 海流速度及其相对于AUV的方向用两个角表示: 迎角 α_c 和侧滑角 β_c 。方程 (33) 中的模型是一个三维海流模型。海流速度通常被建模为具有高斯白噪声输入 η 的一阶系统 [56]:

$$\dot{U}_c + \zeta U_c = \mu \quad (34)$$

我们选择 $\zeta \geq 0$, 使得方程 (34) 导致一个稳定的系统。除了 U_c , 我们还可以将动力学关联到 α_c 和 β_c 。

3 Level Flight Dynamics and Linearization

3.1 Level flight dynamics

平飞是指在没有任何干扰的情况下, 车辆接受一组恒定输入, 并保持固定速度和方向的状态。这也被称为 *trim* 状态。这是一个关键过程, 涉及调整车辆的配置和控制设置, 以在水中实现平衡和稳定状态。当自主水下航行器 (AUV) 位于水面以下时, 会同时受到许多力的作用。为了使AUV满足前面提到的平飞条件, 所有这些力矩都需要保持平衡[57], 这将进一步导致三轴上的加速度和旋转速率为零。从数学上讲, 这就成为了一个估计控制参数以实现该状态的问题。这里的控制面包括尾舵角、方向舵角和螺旋桨。

$$\mathbf{u}_i = [n \ \delta_{s_1} \ \delta_{s_2} \ \delta_{r_1} \ \delta_{r_2}]^T$$

上述向量由螺旋桨转速、两个船尾角和两个舵角组成。为了求解修整, 车辆的数学模型被实现为一个函数, 该函数接受车辆状态和控制输入, 并给出所有变化率。

the vehicle states as output. That means except horizontal position all other states will have their derivatives zero.

$$\Omega = [\dot{U} \ \dot{\alpha} \ \dot{\beta} \ \dot{p} \ \dot{q} \ \dot{r} \ \dot{z} \ \dot{\phi} \ \dot{\theta} \ \dot{\psi} \ \dot{n} \ u_p]^T$$

Here U is the speed of the vehicle expressed as $\sqrt{u^2 + v^2 + w^2}$, α is the angle of attack, β is the side-slip angle of the vehicle and u_p is the propeller inflow velocity. Since the vehicle is not rotating, the body frame rotation rates are zero, i.e. $\dot{\phi} = \dot{\theta} = \dot{\psi} = p = q = r = 0$. This implies the states in Ω should become zero. To achieve this optimum variables are chosen to be decision variables. In this case, the variables are,

$$\mathbf{d} = [\alpha \ \beta \ \theta \ n \ u_p \ \tau \ \delta_{s_1} \ \delta_{s_2} \ \delta_{r_1} \ \delta_{r_2}]^T$$

There are two ways of estimating these trim states. One is using Numerical methods like Newton Raphson. Another way is an analytical method wherein we put certain assumptions and based on that estimate expressions for all decision variables. After putting conditions the reduced 6 DOF equations will become

$$(B - W)\theta + X_{u|u}|U^2 + (1 - \tau_p) T_{n|n}|n^2 = 0 \quad (35)$$

$$(W - B)\phi + Y_{uv}Uv + Y_{v|v}|v^2 + Y_{uu\delta_r}U^2(\delta_{r_1} + \delta_{r_2}) = 0 \quad (36)$$

$$(W - B)[1 - 0.5\theta^2 - 0.5\phi^2] - Z_{w|w}|U^2\theta^2 + Z_{uw}U\theta + Z_{uu\delta_s}U^2(\delta_{s_1} + \delta_{s_2}) = 0 \quad (37)$$

$$-z_g W\phi + Q_{n|n}|n^2 + K_{roll}\delta_{roll} = 0 \quad (38)$$

$$z_g(W + M_{uw}U^2)\theta - M_{w|w}|U^2\theta^2 + M_{uu\delta_s}U^2(\delta_{s_1} + \delta_{s_2}) = 0 \quad (39)$$

$$N_{uv}Uv + N_{v|v}|v^2 + N_{uu\delta_r}U^2(\delta_{r_1} + \delta_{r_2}) = 0 \quad (40)$$

Furthermore, we address the trim scenario, wherein the objective is to nullify the roll motion. This can be effectively achieved through the utilization of decoupled control surfaces. By independently manipulating the two sterns ($\delta_{s_1}, \delta_{s_2}$) and two rudders ($\delta_{r_1}, \delta_{r_2}$), distinct adjustments can be made to ensure that the roll angle ϕ is minimized. Consequently, this configuration results in five essential control inputs to maintain a level flight condition: four control surface settings ($\delta_{s_1}, \delta_{s_2}, \delta_{r_1}$, and δ_{r_2}), alongside a singular RPM value (n_{prop}).

3.2 Linearization

Linearization of an AUV involves approximating its nonlinear dynamic model with a simplified linear model around a specific operating point or reference condition. This linearization process is often used for control system design, stability analysis, and controller tuning. We consider linearization in two planes: one in the vertical plane known as the depth plane or dive plane, while the other is the horizontal plane known as the heading or steering subsystem. We would These two subsystems involve approximating the nonlinear dynamic behaviour of the AUV's depth (vertical motion) and heading (yaw rotation) around a specific operating speed U .

Depth - Pitch Subsystem: In the depth plane, linearization focuses on understanding how changes in control surfaces δ_{s_1} and δ_{s_2} , impact the AUV's vertical position. Since sterns are identical we have $\delta_s = \delta_{s_1} + \delta_{s_2}$ assuming no change in roll ϕ . We consider the motion of the vehicle in the vertical plane with body-relative heave velocity w , the pitch rate q , the earth-relative vehicle depth z , and pitch angle θ . Setting to zero the all unrelated terms (v, p, r, ϕ, y_g, z_g) in eq. (7), eq. (10), eq. (11),

车辆状态作为输出。这意味着除了水平位置之外，所有其他状态的导数都为零。

$$\Omega = [\dot{U} \ \dot{\alpha} \ \dot{\beta} \ \dot{p} \ \dot{q} \ \dot{r} \ \dot{z} \ \dot{\phi} \ \dot{\theta} \ \dot{\psi} \ \dot{n} \ u_p]^T$$

这里 U 是以 $\sqrt{u^2 + v^2 + w^2}$ 表示的车辆速度， α 是攻角， β 是车辆侧滑角， u_p 是螺旋桨入流速度。由于车辆不旋转，机体坐标系的旋转速率为零，即 $\dot{\phi} = \dot{\theta} = \dot{\psi} = p = q = r = 0$ 。这意味着 Ω 中的状态应该变为零。为了实现这一点，最优变量被选择为决策变量。在这种情况下，变量为，

$$\mathbf{d} = [\alpha \ \beta \ \theta \ n \ u_p \ \tau \ \delta_{s_1} \ \delta_{s_2} \ \delta_{r_1} \ \delta_{r_2}]^T$$

估计这些整定状态有两种方法。一种是使用数值方法，如牛顿-拉夫森方法。另一种方法是解析方法，在这种方法中，我们提出某些假设，并基于这些假设估算所有决策变量的表达式。在施加条件后，化简的六自由度方程将变为

$$(B - W)\theta + X_{u|u}|U|^2 + (1 - \tau_p) T_{n|n}|n|^2 = 0 \quad (35)$$

$$(W - B)\phi + Y_{uv}Uv + Y_{v|v}|v|^2 + Y_{uu\delta_r}U^2(\delta_{r_1} + \delta_{r_2}) = 0 \quad (36)$$

$$(W - B)[1 - 0.5\theta^2 - 0.5\phi^2] - Z_{w|w}|U|^2\theta^2 + Z_{uw}U\theta + Z_{uu\delta_s}U^2(\delta_{s_1} + \delta_{s_2}) = 0 \quad (37)$$

$$-z_g W\phi + Q_{n|n}|n|^2 + K_{roll}\delta_{roll} = 0 \quad (38)$$

$$z_g(W + M_{uw}U^2)\theta - M_{w|w}|U|^2\theta^2 + M_{uu\delta_s}U^2(\delta_{s_1} + \delta_{s_2}) = 0 \quad (39)$$

$$N_{uv}Uv + N_{v|v}|v|^2 + N_{uu\delta_r}U^2(\delta_{r_1} + \delta_{r_2}) = 0 \quad (40)$$

此外，我们处理修剪情景，其目标是消除滚转运动。这可以通过使用解耦控制面有效实现。通过独立操作两个船尾 ($\delta_{s_1}, \delta_{s_2}$) 和两个舵 ($\delta_{r_1}, \delta_{r_2}$)，可以进行不同的调整，以确保滚转角 ϕ 最小化。因此，这种配置产生五个基本控制输入以维持水平飞行状态：四个控制面设置 (δ_{s_1} 、 δ_{s_2} 、 δ_{r_1} 和 δ_{r_2}) 以及一个单一的转速值 (n_{prop})。

3.2 Linearization

AUV 的线性化涉及将其非线性动态模型在特定工作点或参考条件下近似为简化的线性模型。该线性化过程通常用于控制系统设计、稳定性分析和控制器调节。我们考虑在两个平面上的线性化：一个是垂直平面，也称为深度平面或潜航平面，另一个是水平平面，也称为航向或转向子系统。这两个子系统涉及在特定工作速度 U 下近似 AUV 的深度（垂直运动）和航向（偏航旋转）的非线性动态行为。

Depth - Pitch Subsystem: 在纵深平面上，线性化集中于理解控制面 δ_{s_1} 和 δ_{s_2} 的变化如何影响 AUV 的垂直位置。由于尾部是相同的，我们有 $\delta_s = \delta_{s_1} + \delta_{s_2}$ 假设滚转 ϕ 没有变化。我们考虑车辆在垂直平面上的运动，具有相对于机体的升沉速度 w 、俯仰角速度 q 、相对于地面的车辆深度 z ，以及俯仰角 θ 。在方程 (7)、方程 (10)、方程 (11) 中，将所有无关项 (v, p, r, ϕ, y_g, z_g) 置零，

eq. (12) and eq. (13), we get

$$\dot{z} = -\sin \theta u + \cos \theta w \quad (41)$$

$$\dot{\theta} = q \quad (42)$$

$$m(\dot{w} - uq - z_g \dot{q}^2 - x_g \dot{q}) = Z \quad (43)$$

$$I_{yy} \dot{q} + m[z_g(\dot{u} + wq) - x_g(\dot{w} - uq)] = M \quad (44)$$

where,

$$\begin{aligned} Z &= Z_{HS} + Z_{w|w}|w| + Z_{q|q}|q| + Z_{\dot{w}}\dot{w} + Z_{\dot{q}}\dot{q} + Z_{uq}uq + Z_{uw}uw + Z_{\delta_s}\delta_s \\ M &= M_{HS} + M_{w|w}|w| + M_{q|q}|q| + M_{\dot{w}}\dot{w} + M_{\dot{q}}\dot{q} + M_{uq}uq + M_{uw}uw + M_{\delta_s}\delta_s \end{aligned} \quad (45)$$

These equations can be linearized by assuming the vehicle perturbations around a steady point U . The speed U in this case represents the steady-state forward velocity of the vehicle; the pitch is linearized about zero. As heave velocity is usually small we set $w = 0$. Finally, substituting $u = U$ and linearized velocities and dropping higher-order terms [36] results in the following linearized equations of motion,

$$\begin{aligned} \dot{z} &= -\theta U + w \\ \dot{\theta} &= q \\ (I_{yy} - M_{\dot{q}})\dot{q} + (mx_g U - M_q)q - M_{\theta}\theta &= M_{\delta_s}\delta_s \end{aligned} \quad (46)$$

where $M_{\dot{q}}$ is the added mass and the linearized coefficients are given as follows,

$$\begin{aligned} M_{\theta} &= -z_g W \\ M_{\delta_s} &= M_{uu\delta_s} U^2 \\ M_q &= M_{uq} U \end{aligned} \quad (47)$$

and $M_{uq} = M_{uqa} + M_{uqf}$ is the combined term with respect to added mass and fin lift coefficient. The equations of motion can be further written down in matrix form in terms of a **three**-state vector containing (q, θ, z) . Assuming that the heave velocities are small compared to the other terms, we can further reduce the equations of motion to this form:

$$\dot{\mathbf{x}} = \mathbf{A}_D \mathbf{x} + \mathbf{B}_D \delta_s \quad (48)$$

where,

$$\mathbf{x} = \begin{bmatrix} q \\ \theta \\ z \end{bmatrix} \quad \mathbf{A}_D = \begin{bmatrix} \frac{M_q}{I_{yy} - M_{\dot{q}}} & -\frac{M_{\theta}}{I_{yy} - M_{\dot{q}}} & 0 \\ 1 & 0 & 0 \\ 0 & -U & 0 \end{bmatrix} \quad \mathbf{B}_D = \begin{bmatrix} \frac{M_{\delta_s}}{I_{yy} - M_{\dot{q}}} \\ 0 \\ 0 \end{bmatrix}$$

The transfer function of the vertical plane subsystem with input δ_s and output z is,

$$G_z(s) \triangleq \frac{z(s)}{\delta_s(s)} = -\frac{U}{s} \frac{\frac{M_{\delta_s}}{I_{yy} - M_{\dot{q}}}}{\left(s^2 - \frac{M_q}{I_{yy} - M_{\dot{q}}}s - \frac{M_{\theta}}{I_{yy} - M_{\dot{q}}}\right)} \quad (49)$$

Further, the inner pitch loop transfer function relating the stern plane angle and pitch can be expressed as,

$$G_{\theta}(s) \triangleq \frac{\theta(s)}{\delta_s(s)} = \frac{\frac{M_{\delta_s}}{I_{yy} - M_{\dot{q}}}}{\left(s^2 - \frac{M_q}{I_{yy} - M_{\dot{q}}}s - \frac{M_{\theta}}{I_{yy} - M_{\dot{q}}}\right)} \quad (50)$$

等式 (12) 和等式 (13), 得到

$$\dot{z} = -\sin \theta u + \cos \theta w \quad (41)$$

$$\dot{\theta} = q \quad (42)$$

$$m(\dot{w} - uq - z_g q^2 - x_g \dot{q}) = Z \quad (43)$$

$$I_{yy} \dot{q} + m[z_g(\dot{u} + wq) - x_g(\dot{w} - uq)] = M \quad (44)$$

哪里,

$$\begin{aligned} Z &= Z_{HS} + Z_{w|w}|w| + Z_{q|q}|q| + Z_{\dot{w}}\dot{w} + Z_{\dot{q}}\dot{q} + Z_{uq}uq + Z_{uw}uw + Z_{\delta_s}\delta_s \\ M &= M_{HS} + M_{w|w}|w| + M_{q|q}|q| + M_{\dot{w}}\dot{w} + M_{\dot{q}}\dot{q} + M_{uq}uq + M_{uw}uw + M_{\delta_s}\delta_s \end{aligned} \quad (45)$$

通过假设车辆在稳态点 U 附近的扰动, 可以将这些方程线性化。在这种情况下, 速度 U 表示车辆的稳态前进速度; 俯仰角围绕零线性化。由于升沉速度通常较小, 我们设 $w = 0$ 。最后, 代入 $u = U$ 和线性化速度并舍弃高阶项 [36], 得到以下线性化运动方程,

$$\begin{aligned} \dot{z} &= -\theta U + w \\ \dot{\theta} &= q \\ (I_{yy} - M_{\dot{q}})\dot{q} + (mx_g U - M_q)q - M_{\theta}\theta &= M_{\delta_s}\delta_s \end{aligned} \quad (46)$$

其中 $M_{\dot{q}}$ 是附加质量, 线性化系数如下所示,

$$\begin{aligned} M_{\theta} &= -z_g W \\ M_{\delta_s} &= M_{uu\delta_s} U^2 \\ M_q &= M_{uq} U \end{aligned} \quad (47)$$

并且 $M_{uq} = M_{uqa} + M_{uqf}$ 是关于附加质量和鳍升力系数的组合项。运动方程可以进一步以包含 (q, θ, z) 的三状态向量的矩阵形式写出。假设纵向速度与其他项相比很小, 我们可以进一步将运动方程简化为这种形式:

$$\dot{\mathbf{x}} = \mathbf{A}_D \mathbf{x} + \mathbf{B}_D \delta_s \quad (48)$$

哪里,

$$\mathbf{x} = \begin{bmatrix} q \\ \theta \\ z \end{bmatrix} \quad \mathbf{A}_D = \begin{bmatrix} \frac{M_q}{I_{yy} - M_{\dot{q}}} & -\frac{M_{\theta}}{I_{yy} - M_{\dot{q}}} & 0 \\ 1 & 0 & 0 \\ 0 & -U & 0 \end{bmatrix} \quad \mathbf{B}_D = \begin{bmatrix} \frac{M_{\delta_s}}{I_{yy} - M_{\dot{q}}} \\ 0 \\ 0 \end{bmatrix}$$

具有输入 δ_s 和输出 z 的垂直平面子系统的传递函数为,

$$G_z(s) \triangleq \frac{z(s)}{\delta_s(s)} = -\frac{U}{s} \frac{\frac{M_{\delta_s}}{I_{yy} - M_{\dot{q}}}}{\left(s^2 - \frac{M_q}{I_{yy} - M_{\dot{q}}}s - \frac{M_{\theta}}{I_{yy} - M_{\dot{q}}}\right)} \quad (49)$$

此外, 描述船尾平面角与俯仰之间关系的内俯仰环传递函数可以表示为,

$$G_{\theta}(s) \triangleq \frac{\theta(s)}{\delta_s(s)} = \frac{\frac{M_{\delta_s}}{I_{yy} - M_{\dot{q}}}}{\left(s^2 - \frac{M_q}{I_{yy} - M_{\dot{q}}}s - \frac{M_{\theta}}{I_{yy} - M_{\dot{q}}}\right)} \quad (50)$$

and the outer loop depth transfer function that relates vehicle pitch θ and the depth z is,

$$G_z(s) \triangleq \frac{z(s)}{\theta(s)} = -\frac{U}{s} \quad (51)$$

Heading/Yaw Subsystem: In the heading subsystem, linearization involves approximating the effects of control surfaces on the AUV's yaw rotation. This is crucial for maintaining a desired heading or direction of travel. Linearized models in the heading plane allow for the design of control algorithms that enable the AUV to accurately follow desired paths, perform turns, or adjust its orientation in response to external factors. Automatic steering or heading control can be done using a rudder δ_{r_1} and δ_{r_2} . Since sterns are identical we have $\delta_r = \delta_{r_1} + \delta_{r_2}$.

The motion of the vehicle in the horizontal plane is considered. The body-relative sway velocity v , the yaw rate r , the earth-relative yaw angle ψ . All other velocities (w, p, q) are set to zero linearize around the operating speed U and substituting $u = U$. We set to zero all unrelated terms (w, p, q, y_g) in eq. (10), eq. (11), eq. (12) and eq. (13). We obtain the following equations,

$$\begin{aligned} \dot{\psi} &= r \\ m(\dot{v} + ur + x_g \dot{r}) &= Y \\ I_{zz} \dot{r} + m[x_g(\dot{v} + ur)] &= N \end{aligned} \quad (52)$$

Substituting $u = U$ and linearized velocities, and coefficients results in the following equations of motion,

$$\begin{aligned} \dot{\psi} &= r \\ (m - Y_{\dot{v}})\dot{v} + (mx_g - Y_{\dot{r}})\dot{r} - Y_{uv}Uv - (Y_{ur} - m)Ur &= Y_{uu}\delta_r U^2 \\ (mx_g - N_{\dot{v}})\dot{v} + (I_{zz} - N_{\dot{r}})\dot{r} - N_{uv}Uv - N_{ur}Ur &= N_{uu}\delta_r U^2 \end{aligned} \quad (53)$$

Further, we assume the sway velocity is small $v = \dot{v} = 0$

$$\begin{aligned} \dot{\psi} &= r \\ (I_{zz} - N_{\dot{r}})\dot{r} - (N_r - mx_g U)r &= N_{\delta_r} \delta_r \end{aligned} \quad (54)$$

The corresponding two state (r, ψ) state-space representation,

$$\dot{\mathbf{x}} = \mathbf{A}_H \mathbf{x} + \mathbf{B}_H \delta_R \quad (55)$$

where,

$$\mathbf{x} = \begin{bmatrix} r \\ \psi \end{bmatrix} \quad \mathbf{A}_H = \begin{bmatrix} \frac{N_r}{I_{zz} - N_{\dot{r}}} & 0 \\ 1 & 0 \end{bmatrix} \quad \mathbf{B}_H = \begin{bmatrix} \frac{N_{\delta_r}}{I_{zz} - N_{\dot{r}}} \\ 0 \end{bmatrix}$$

The transfer function with δ_r as input and ψ as output is given by,

$$G_\psi(s) \triangleq \frac{\psi(s)}{\delta_r(s)} = \frac{\frac{N_{\delta_r}}{I_{zz} - N_{\dot{r}}}}{\left(s^2 - \frac{N_r}{I_{zz} - N_{\dot{r}}}\right)} \quad (56)$$

Further, we have the following relationship between y and ψ ,

$$\dot{y} = U\psi \quad (57)$$

Hence the transfer function,

$$G_y(s) \triangleq \frac{y(s)}{\delta_r(s)} = \frac{U}{s} \frac{\frac{N_{\delta_r}}{I_{zz} - N_{\dot{r}}}}{s^2 - \frac{N_r}{I_{zz} - N_{\dot{r}}}} \quad (58)$$

where $N_{\dot{r}}$ and N_{δ_r} are respectively the added mass and the fin lift coefficient.

以及将车辆纵倾角 θ 与深度 z 相关联的外环深度传递函数为,

$$G_z(s) \triangleq \frac{z(s)}{\theta(s)} = -\frac{U}{s} \quad (51)$$

Heading/Yaw Subsystem: 在航向子系统中, 线性化涉及对控制舵面对水下自主航行器 (AUV) 偏航旋转的影响进行近似。这对于维持所需的航向或行驶方向至关重要。航向平面中的线性化模型允许设计控制算法, 使AUV能够准确地沿预定路径行驶、执行转向或根据外部因素调整其方向。自动转向或航向控制可以使用舵 δ_{r_1} 和 δ_{r_2} 完成。由于尾部是相同的, 我们有 $\delta_r = \delta_{r_1} + \delta_{r_2}$ 。

考虑车辆在水平面上的运动。车体相关的横摆速度 v 、偏航率 r 、相对于地面的偏航角 ψ 。所有其他速度 (w, p, q) 被设为零, 在工作速度 U 附近线性化并代入 $u = U$ 。我们在方程 (10)、方程 (11)、方程 (12) 和方程 (13) 中将所有无关项 (w, p, q, y_g) 设为零。我们得到以下方程,

$$\begin{aligned} \dot{\psi} &= r \\ m(\dot{v} + ur + x_g \dot{r}) &= Y \\ I_{zz} \dot{r} + m[x_g(\dot{v} + ur)] &= N \end{aligned} \quad (52)$$

代入 $u = U$ 以及线性化速度和系数, 得到以下运动方程,

$$\begin{aligned} \dot{\psi} &= r \\ (m - Y_{\dot{v}})\dot{v} + (mx_g - Y_{\dot{r}})\dot{r} - Y_{uv}Uv - (Y_{ur} - m)Ur &= Y_{uu}\delta_r U^2 \\ (mx_g - N_{\dot{v}})\dot{v} + (I_{zz} - N_{\dot{r}})\dot{r} - N_{uv}Uv - N_{ur}Ur &= N_{uu}\delta_r U^2 \end{aligned} \quad (53)$$

此外, 我们假设摆动速度很小 $v = \dot{v} =$ 零

$$\begin{aligned} \dot{\psi} &= r \\ (I_{zz} - N_{\dot{r}})\dot{r} - (N_r - mx_g U)r &= N_{\delta_r} \delta_r \end{aligned} \quad (54)$$

对应的二维 (r, ψ) 状态空间表示,

$$\dot{\mathbf{x}} = \mathbf{A}_H \mathbf{x} + \mathbf{B}_H \delta_R \quad (55)$$

哪里,

$$\mathbf{x} = \begin{bmatrix} r \\ \psi \end{bmatrix} \quad \mathbf{A}_H = \begin{bmatrix} \frac{N_r}{I_{zz} - N_{\dot{r}}} & 0 \\ 1 & 0 \end{bmatrix} \quad \mathbf{B}_H = \begin{bmatrix} \frac{N_{\delta_r}}{I_{zz} - N_{\dot{r}}} \\ 0 \end{bmatrix}$$

以 δ_r 为输入、 ψ 为输出的传递函数表示为,

$$G_\psi(s) \triangleq \frac{\psi(s)}{\delta_r(s)} = \frac{\frac{N_{\delta_r}}{I_{zz} - N_{\dot{r}}}}{\left(s^2 - \frac{N_r}{I_{zz} - N_{\dot{r}}}\right)} \quad (56)$$

此外, 我们有 y 和 ψ 之间的以下关系,

$$\dot{y} = U\psi \quad (57)$$

因此, 传递函数为,

$$G_y(s) \triangleq \frac{y(s)}{\delta_r(s)} = \frac{U}{s} \frac{\frac{N_{\delta_r}}{I_{zz} - N_{\dot{r}}}}{s^2 - \frac{N_r}{I_{zz} - N_{\dot{r}}}} \quad (58)$$

在哪里 $N_{\dot{r}}$ 并且 N_{δ_r} 分别是附加质量和鳍翼升力系数

Speed Subsystem Only the surge equation of motion is considered to simplify the speed subsystem. We assume that the sway, heave, pitch, roll, and yaw coupling is insignificant. Consequently, the surge equation can be simplified by merely retaining the terms referring to acceleration $\dot{u}$. We will set to zero all other velocities (v, w, p, q, r), and drop the equations for any other direction.

$$(m - X_{\dot{u}})\dot{u} = X_{u|u}|u| + X_{prop} \quad (59)$$

where $X_{u|u}$ is the axial drag and X_{prop} is the thrust force due to the propeller in the X-direction. Using trigonometric small-angle approximation, we arrive at the following result with U as the surge velocity of the vehicle and substituting linearized axial drag X_u and X_{prop} , we get:

$$(m - X_{\dot{u}})\dot{u} = X_u u + (1 - \tau)T_{nn}n|n| \quad (60)$$

where n is the angular velocity of the propeller. This equation is non-linear, so we need to linearize the $\bar{n}^2$ term. This is accomplished by fitting a slope to the parabolic angular-velocity curve.

$$n^2 = m_n n$$

with m_n is a coefficient of the polynomial. The transfer function for the speed subsystem,

$$G_u(s) \triangleq \frac{u(s)}{n(s)} = \frac{m_n(1 - \tau)T_{nn}}{(m - X_{\dot{u}})s - X_u} \quad (61)$$

We develop the control architecture in the preceding section for the horizontal plane (Heading), vertical plane (Depth - pitch) and speed subsystems.

4 Simulation Results

Control systems are paramount in ensuring the AUV's autonomy and mission success. The design of control algorithms relies on the mathematical model of the AUV to determine appropriate control inputs that steer the vehicle to desired positions or trajectories. These control systems can be based on classical control techniques or modern approaches such as conventional controllers or SMC.

4.1 Linear Controllers

This section presents a comprehensive study of the linear controller architecture for the three subsystems: horizontal plane, vertical plane, and speed subsystem. The simulation results for these are also presented.

Depth-pitch controller: The depth control system of a linearized AUV is a critical component responsible for maintaining the desired depth and pitch orientation during its underwater operations. The schematic of the depth control system consists of an outer depth loop and an inner pitch loop as shown in Figure 3. The transfer function of the subsystem with U as 4 knots is,

$$G_z(s) = \frac{4.154}{s(s^2 + 0.824s + 0.6927)} \quad (62)$$

For the outer depth loop, the required depth command is passed through a first-order low-pass filter to facilitate the refinement and smoothness of the curve. Eventually, the error between the

Speed Subsystem 为了简化速度子系统，仅考虑纵摇运动方程。我们假设横摇、纵振、俯仰、横滚和偏航耦合可以忽略。因此，纵摇方程可以通过仅保留与加速度 $\dot{u}$ 相关的项来简化。我们将所有其他速度 (v, w, p, q, r) 设为零，并删除任何其他方向的方程。

$$(m - X_{\dot{u}})\dot{u} = X_{u|u}|u| + X_{prop} \quad (59)$$

其中 $X_{u|u}$ 是轴向阻力， X_{prop} 是由螺旋桨在 X 方向产生的推力。使用三角函数小角度近似，我们得到以下结果，其中 U 是车辆的冲程速度，并代入线性化的轴向阻力 X_u 和 X_{prop} ，我们得到：

$$(m - X_{\dot{u}})\dot{u} = X_u u + (1 - \tau)T_{nn}n|n| \quad (60)$$

其中 n 是螺旋桨的角速度。这个方程是非线性的，因此我们需要线性化 $\bar{n}^2$ 项。这可以通过对抛物线形的角速度曲线拟合斜率来实现。

$$n^2 = m_n n$$

其中 m_n 是多项式的系数。速度子系统的传递函数，

$$G_u(s) \triangleq \frac{u(s)}{n(s)} = \frac{m_n(1 - \tau)T_{nn}}{(m - X_{\dot{u}})s - X_u} \quad (61)$$

我们在前一节中为水平平面（航向）、垂直平面（深度 - 俯仰）和速度子系统开发了控制架构。

4 Simulation Results

控制系统在确保AUV自主性和任务成功中至关重要。控制算法的设计依赖于AUV的数学模型，以确定适当的控制输入，将车辆引导到期望的位置或轨迹。这些控制系统可以基于经典控制技术或现代方法，例如传统控制器或滑模控制（SMC）。

4.1 Linear Controllers

本节介绍了三大子系统的线性控制器结构的全面研究：水平平面、垂直平面和速度子系统。同时也展示了这些的仿真结果。

Depth-pitch controller: 线性化AUV的深度控制系统是一个关键组件，负责在水下操作过程中保持所需的深度和俯仰姿态。深度控制系统的示意图由一个外部深度回路和一个内部俯仰回路组成，如图3所示。以4节的 U 为基准的子系统传递函数为，

$$G_z(s) = \frac{4.154}{s(s^2 + 0.824s + 0.6927)} \quad (62)$$

对于外层深度循环，所需的深度指令通过一阶低通滤波器，以便于曲线的优化和平滑。最终，

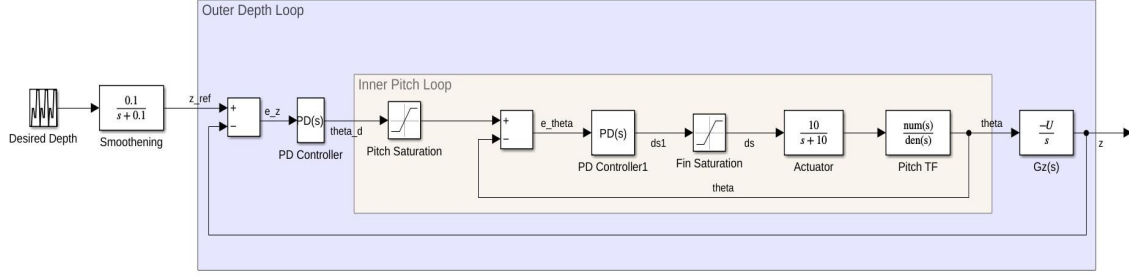


Fig. 3: Schematic of Depth - pitch subsystem with controller

desired depth and the actual depth is fed to a Proportional-Derivative (PD) controller which gives the desired pitch angle as output. The outer depth loop provides the desired pitch angle θ_{ref} and the control law is,

$$\theta_{ref} = - \left(K_{pd}e_z + K_{dd}\frac{d}{dt}e_z \right) \quad (63)$$

where $e_z = z_{ref} - z$ error between the desired (z_{ref}) and actual depth (z), K_{pd} and K_{id} are the inner loop proportional and derivative gains respectively. Subsequently, the PD controller takes the error between this desired pitch angle and the actual pitch angle as input and outputs the required stern angle command. The control law for the inner pitch loop can be expressed as:

$$\delta_s = - \left(K_{pp}e_\theta + K_{dp}\frac{d}{dt}e_\theta \right) \quad (64)$$

where δ_s is the required stern angle and $e_\theta = \theta_{ref} - \theta$ is the error between the required pitch angle θ_{ref} and actual pitch angle θ , K_{pp} and K_{dp} are the outer loop proportional and derivative gains respectively. The simulation results of the depth-pitch subsystem with the controller are shown in Figure 4. The reference signal for the desired depth (z_{ref}) is depicted as a solid red line and serves as the input reference for the system. The first subplot presents a comparative analysis of the desired and actual depth profiles over a specified time span. This visual comparison provides valuable insights into the depth control performance of the controller, illustrating how effectively the controller responds to dynamic changes in depth references. Moving to the second subplot, a clear representation of the vehicle's pitch angle response within the influence of the applied control system is observed. The plots of the θ_{ref} and θ over time indicate that the controller accurately tracks the desired pitch angle, showcasing its precision. The third subplot, depicted in Figure 4, illustrates the evolution of the required stern angle command over time in response to varying desired depth commands. Notably, due to the small magnitude of the depth change, the maximum required stern angle remains around 5 degrees. This analysis effectively demonstrates the controller's ability to accurately follow the desired depth trajectory.

Yaw Controller: We follow a similar architecture for yaw control of the vehicle. The schematic of the yaw control is shown in Figure 5. There are two loops as before, the inner yaw loop and the outer y direction control loop. The transfer function of the yaw subsystem is,

$$G_\psi(s) = -\frac{4.223}{s(s+0.579)} \quad (65)$$

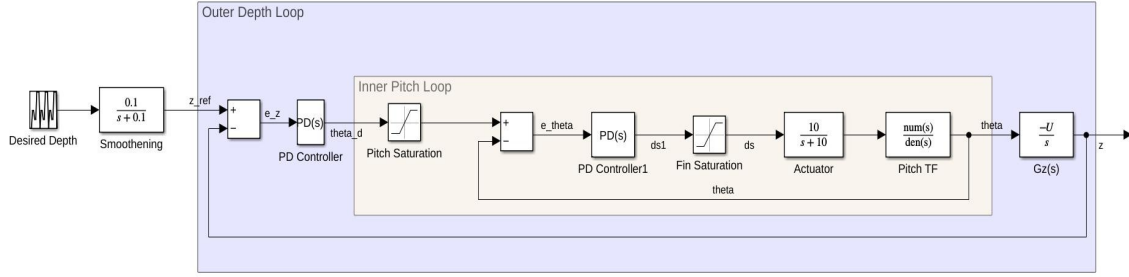


图3: 带控制器的深度-俯仰子系统示意图

所需深度与实际深度被送入比例-微分 (PD) 控制器, 输出所需的俯仰角。外部深度环路提供所需的俯仰角 θ_{ref} , 控制律为,

$$\theta_{ref} = - \left(K_{pd}e_z + K_{dd}\frac{d}{dt}e_z \right) \quad (63)$$

$e_z = z_{ref} - z$ 是期望深度 (z_{ref}) 与实际深度 (z) 之间的误差, K_{pd} 和 K_{id} 分别为内环的比例增益和微分增益。随后, PD 控制器以该期望俯仰角与实际俯仰角之间的误差为输入, 并输出所需的舵角指令。内俯仰环的控制律可以表示为:

$$\delta_s = - \left(K_{pp}e_\theta + K_{dp}\frac{d}{dt}e_\theta \right) \quad (64)$$

其中 δ_s 是所需的艇尾角, $e_\theta = \theta_{ref} - \theta$ 是所需俯仰角 θ_{ref} 与实际俯仰角 θ 之间的误差, K_{pp} 和 K_{dp} 分别是外环比例和微分增益。带有控制器的深度-俯仰子系统的仿真结果如图 4 所示。所需深度 (z_{ref}) 的参考信号以红色实线表示, 并作为系统的输入参考。第一个子图展示了在特定时间范围内所需深度和实际深度曲线的比较分析。这个可视化比较为控制器的深度控制性能提供了宝贵的见解, 说明控制器对于深度参考的动态变化响应的效果。接着, 在第二个子图中, 可以清楚地看到车辆在所施加控制系统影响下的俯仰角响应。随时间变化的 θ_{ref} 和 θ 曲线表明控制器能够准确跟踪

Yaw Controller: 我们遵循类似的架构来控制车辆偏航。偏航控制的示意图如图5所示。和之前一样, 有两个回路, 内部偏航回路和外部y方向控制回路。偏航子系统的传递函数是,

$$G_\psi(s) = -\frac{4.223}{s(s+0.579)} \quad (65)$$

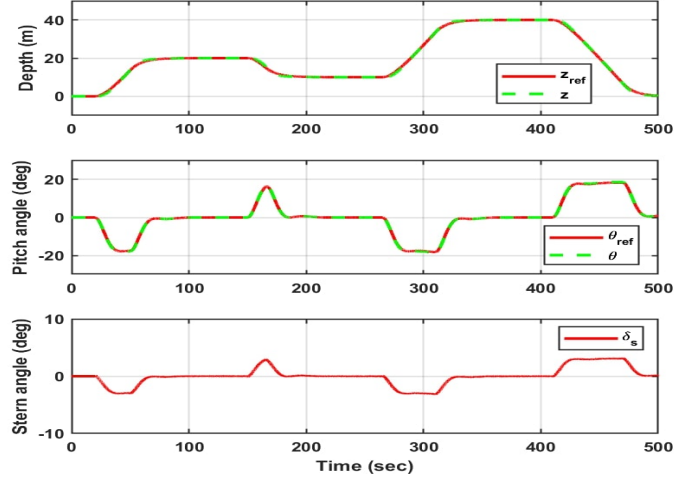


Fig. 4: Depth, pitch and stern angle of depth subsystem

The control law for the inner loop is given by,

$$\delta_r = - \left(K_{p\psi} e_\psi + K_{d\psi} \frac{d}{dt} e_\psi \right) \quad (66)$$

where δ_r is the required rudder angle to provide the required yawing of the vehicle and $e_\psi = \psi_{ref} - \psi$. $K_{p\psi}$ and $K_{d\psi}$ are the inner loop controller gains. The outer loop provides the desired yaw angle ψ_{ref} and the control law is,

$$\psi_{ref} = - \left(K_{py} e_y + K_{dy} \frac{d}{dt} e_y \right) \quad (67)$$

where $e_y = y_{ref} - y$ error between the desired and actual y and K_{py} and K_{dy} are the outer loop controller gains.

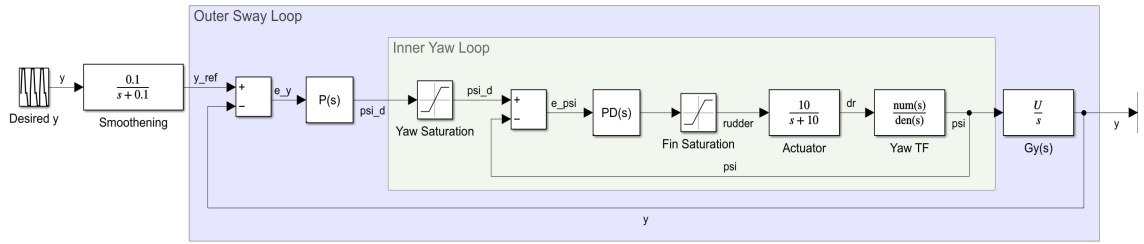


Fig. 5: Schematic of yaw subsystem with controller

In a similar fashion as the depth subsystem, a signal of varying y is sent to the outer controller as a reference and the desired rudder angle is obtained for the same. The results of a similar manoeuvre

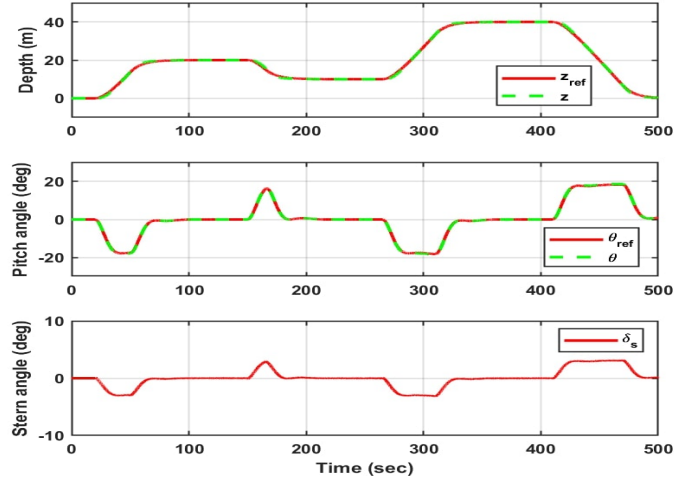


图4: 深度子系统的深度、纵倾角和艉角

内环的控制律如下所示,

$$\delta_r = - \left(K_{p\psi} e_\psi + K_{d\psi} \frac{d}{dt} e_\psi \right) \quad (66)$$

其中 δ_r 是提供车辆所需偏航的必要舵角和 $e_\psi = \psi_{ref} - \psi$ 。 $K_{p\psi}$ 和 $K_{d\psi}$ 是内环控制器增益。外环提供所需偏航角 ψ_{ref} , 控制律为,

$$\psi_{ref} = - \left(K_{py} e_y + K_{dy} \frac{d}{dt} e_y \right) \quad (67)$$

其中 $e_y = y_{ref} - y$ 是期望与实际 y 、 K_{py} 和 K_{dy} 之间的误差, 是外环控制器增益。

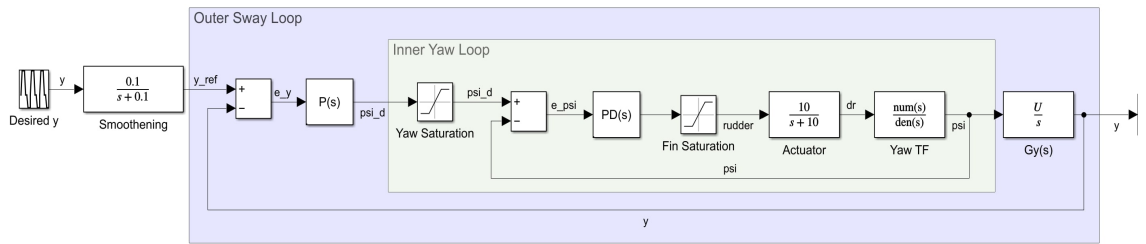


图5: 带控制器的偏航子系统示意图

与深度子系统类似, 将变化的 y 信号发送到外部控制器作为参考, 从而得到所需的舵角。类似机动的结果

are given in Figure 6. Here the first subplot represents the desired lateral position(y_{ref}) and actual lateral position(y). It is seen that the desired position is being followed precisely. The respective yaw angles are given in the second subplot. The maximum value of the yaw angle depends on the value of y_{ref} as well as the time taken to reach the same. As seen from the figure, in this manoeuvre for the demand of zero to 20 meters in around one minute at a constant speed of 3 knots a maximum of 15 degrees of yaw angle is required which is being followed accurately. The third subplot gives the desired rudder angle to obtain the yaw requirement for the given manoeuvre.

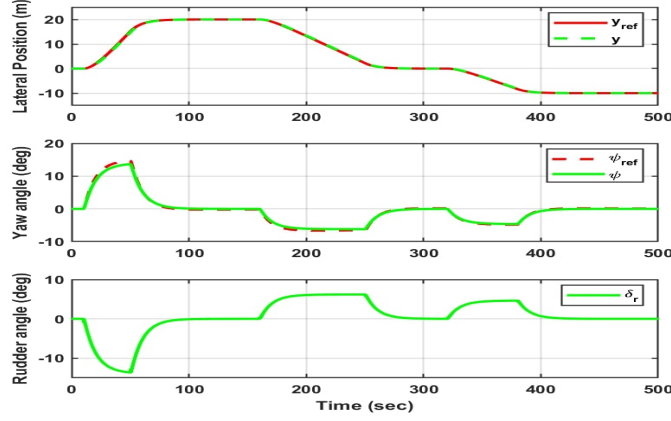


Fig. 6: Lateral position, yaw and rudder angle of yaw subsystem

Speed subsystem: The presented simulation results offer a comprehensive portrayal of the speed controller's tracking capabilities. Through a comparative analysis, the required speed is contrasted with the actual speed profile of the vehicle, spanning a specified time interval. This comprehensive control system architecture is shown in Figure 7, offering a tangible overview of the interactions between the components involved in regulating the AUV's speed. The desired speed command (u_{ref}) is compared with the actual speed (u). The difference between these two values is fed to the Proportional Derivative (PD) controller, which subsequently generates the necessary RPM output to achieve the specified speed. This RPM output then acts as an input to the system, as encapsulated by the transfer function outlined in eq. (61). The transfer function is,

$$G_u(s) = \frac{0.038}{(31.41s + 1.6)} \quad (68)$$

The speed control system is designed by applying the following control law:

$$n_{prop}(t) = K_{ps}e_{spe} + K_{ds}\frac{d}{dt}e_{spe} \quad (69)$$

where $e_u = u_{ref} - u$, the error between the desired speed and the actual speed of the vehicle, K_{ps} and K_{ds} are proportional and derivative gains for speed controller respectively.

In addition to speed tracking response of the vehicle's propulsion system is examined in terms of motor RPM. By examining the change in required RPM values to maintain the desired speed we can

如图6所示。这里第一个子图表示期望横向位置(y_{ref})和实际横向位置(y)。可以看出, 期望位置被精确跟踪。相应的偏航角在第二个子图中给出。偏航角的最大值取决于 y_{ref} 的数值以及达到该数值所需的时间。从图中可以看出, 在这一机动中, 为了在约一分钟内以3节的恒定速度从零到20米的需求, 需要的最大偏航角为15度, 并且这一偏航角被准确跟踪。第三个子图给出了为实现所述机动的偏航要求所需的舵角。

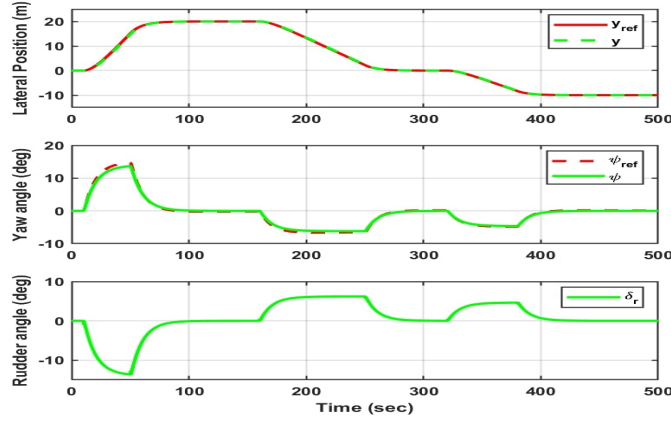


图6: 偏航子系统的横向位置、偏航角和舵角

Speed subsystem: 所呈现的仿真结果提供了速度控制器跟踪能力的全面描绘。通过比较分析, 车辆的所需速度与实际速度曲线在指定时间间隔内进行了对比。该综合控制系统架构如图7所示, 提供了调节AUV速度的各组件之间相互作用的直观概览。期望速度指令 (u_{ref}) 与实际速度 (u) 进行比较。这两个值之间的差值被送入比例微分 (PD) 控制器, 该控制器随后产生实现指定速度所需的RPM输出。该RPM输出随后作为系统的输入, 如公式 (61) 所述的传递函数所体现。传递函数为:

$$G_u(s) = \frac{0.038}{(31.41s + 1.6)} \quad (68)$$

速度控制系统是通过应用以下控制律设计的:

$$n_{prop}(t) = K_{ps}e_{spe} + K_{ds}\frac{d}{dt}e_{spe} \quad (69)$$

其中 $e_u = u_{ref} - u$ 是车辆期望速度与实际速度之间的误差, K_{ps} 和 K_{ds} 分别是速度控制器的比例增益和微分增益。

除了速度跟踪之外, 还会通过电机转速来检查车辆推进系统的响应。通过检查维持所需速度所需转速值的变化, 我们可以

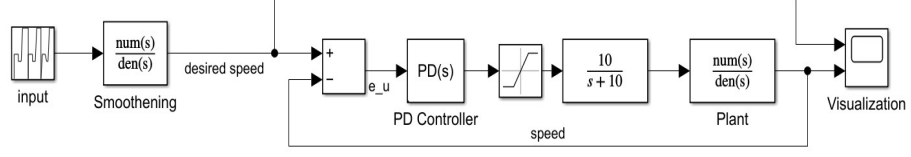


Fig. 7: Schematic of speed subsystem

gauge the control effort generated by the controller and understand the performance of the controller. Figure 8 depicts the outcomes of our linear simulations. The first subplot displays the desired and actual speed profiles, demonstrating the controller's performance in maintaining the desired speed. Here u_{ref} is the desired speed and u is the actual speed attained by the vehicle. The second subplot presents corresponding required RPM values n_{prop} , providing insight into the control effort exerted on the propulsion system. These results collectively showcase the effectiveness and robustness of the speed controller for varying demands.

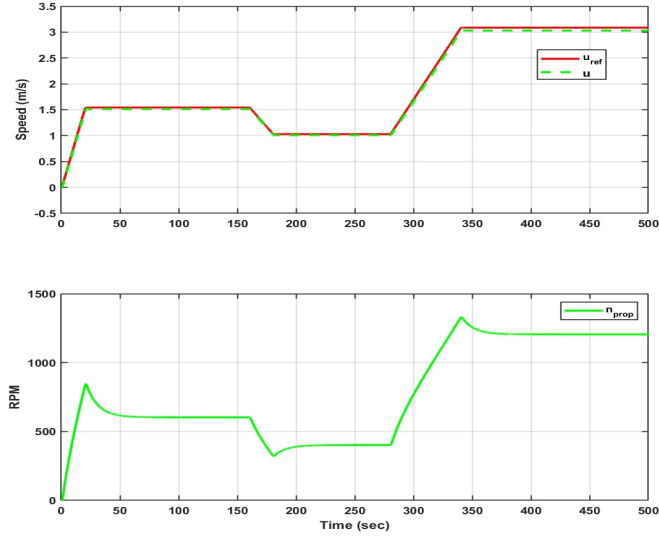


Fig. 8: Speed and RPM of the vehicle

4.2 Level Flight Results

This section presents the outcomes of AUV's level flight analysis. To achieve this, the nonlinear AUV model undergoes linearization at the operational velocity of $U = 4$ knots. The simulation incorporates specific prerequisites: a consistent speed profile ($\dot{u} = \dot{v} = \dot{w} = 0$), constant depth and heading ($z = \psi = \text{constant}$), as well as negligible rotation rates ($\dot{\phi} = \dot{\theta} = \dot{\psi} = 0$) and angular

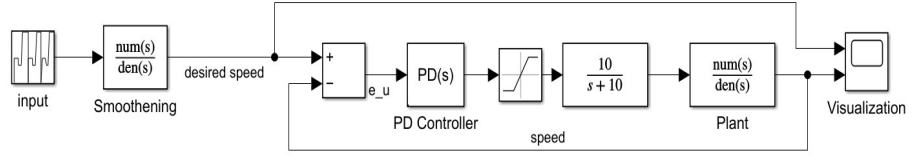


图7：速度子系统示意图

评估控制器产生的控制作用力，并了解控制器的性能。图8显示了我们线性仿真的结果。第一个子图展示了期望速度和实际速度曲线，说明了控制器在维持期望速度方面的性能。其中 u_{ref} 是期望速度， u 是车辆实际达到的速度。第二个子图显示了相应所需的转速值 n_{prop} ，提供了对驱动系统所施加控制作用力的见解。这些结果共同展示了速度控制器在不同需求下的有效性和鲁棒性。

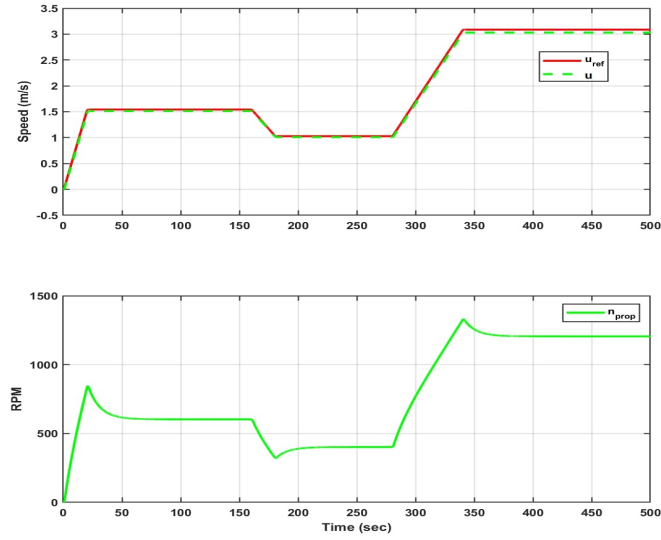


图 8：车辆的速度和转速

4.2 Level Flight Results

本节展示了AUV平飞分析的结果。为实现这一点，非线性AUV模型在运行速度为 $U = 4$ 节时进行了线性化。仿真包含特定前提条件：速度曲线一致 ($\dot{u} = \dot{v} = \dot{w} = 0$)、深度和航向恒定 ($z = \psi = \text{恒定}$)、以及可忽略的旋转率 ($\dot{\phi} = \dot{\theta} = \dot{\psi} = 0$) 和角速度。

velocities ($p = q = r = 0$). The results of the level flight estimation for the research vehicle at 4 knots are summarized in Table 3.

Table 3: Level flight of AUV at 4 Knots

Variable	Symbol	Units	Value
Body-frame surge velocity	u	ms^{-1}	2.0577
Body-frame sway velocity	v	ms^{-1}	0.001
Body-frame heave velocity	w	ms^{-1}	-0.011
Earth-frame roll	ϕ	deg	-2.5941
Earth-frame pitch	θ	deg	-0.72
Propeller rotation rate	n	rpm	1413
Angle of attack	α	deg	-0.72
Angle of sideslip	β	deg	-0.0276
Propeller inflow rate	u_p	ms^{-1}	1.2936
Stern angle 1/ 2	δ_{s1}/δ_{s2}	deg	-1.4
Rudder angle 1/2	$\delta_{r1} / \delta_{r2}$	deg	-0.05

These level flight estimated values were given as initial condition along with the constant control input vector,

$$\begin{bmatrix} n & \delta_{s1} & \delta_{s2} & \delta_{r1} & \delta_{r2} \end{bmatrix}^T = \begin{bmatrix} 1413 & -1.2648 & -1.2648 & -0.0561 & -0.0561 \end{bmatrix}^T \quad (70)$$

The control inputs n are expressed in RPM, while the stern and rudder angles are in degrees. The trim values (shown as a solid red line) are compared against the actuals wherever relevant in Figures 9 and 10. Figure 9 (a) demonstrates that the AUV effectively sustains consistent velocities u , v , and w as given in Table 3. Additionally, the plot in Figure 9 (b) depicts the persistent motion along the x -direction, while exhibiting no discernible variation along the y and z axes. Furthermore, Figure 10 portrays the angular velocities, all of which are maintained at zero. The AUV's roll angle is stabilized at -2.594° , while the pitch remains at -0.72° . Notably, the yaw angle remains constant at 0° . It is evident that the vehicle maintains its steady state for the trim values given in Table 3.

4.3 Nonlinear Controller

The conventional controllers and SMC are implemented to achieve the desired speed, depth and steering. The nonlinear model is given in eq. (14) is implemented in MATLAB-Simulink. The schematic of the research vehicle in Simulink is shown in Figure 11. The References subsystem serves as the source of desired values, encompassing speed, y-position, z-position, and roll angle. These desired values are subsequently transmitted as inputs to the controller block. Within the controller block, there exist four distinct controllers, each aligned with a specific reference signal. Each controller

速度 ($p = q = r = 0$)。研究飞行器在4节飞行速度下平飞估算的结果汇总在表3中。

表3: AUV以4节速度水平飞行

Variable	Symbol	Units	Value
Body-frame surge velocity	u	ms^{-1}	2.0577
Body-frame sway velocity	v	ms^{-1}	0.001
Body-frame heave velocity	w	ms^{-1}	-0.011
Earth-frame roll	ϕ	deg	-2.5941
Earth-frame pitch	θ	deg	-0.72
Propeller rotation rate	n	rpm	1413
Angle of attack	α	deg	-0.72
Angle of sideslip	β	deg	-0.0276
Propeller inflow rate	u_p	ms^{-1}	1.2936
Stern angle 1/ 2	δ_{s1}/δ_{s2}	deg	-1.4
Rudder angle 1/2	$\delta_{r1} / \delta_{r2}$	deg	-0.05

这些平飞的估计值被作为初始条件给出, 并伴随恒定控制输入向量,

$$\begin{bmatrix} n & \delta_{s1} & \delta_{s2} & \delta_{r1} & \delta_{r2} \end{bmatrix}^T = \begin{bmatrix} 1413 & -1.2648 & -1.2648 & -0.0561 & -0.0561 \end{bmatrix}^T \quad (70)$$

控制输入 n 以转速 (RPM) 表示, 而船尾和舵角以度数表示。修整值 (以实心红线显示) 在相关情况下与实际值在图 9 和图 10 中进行比较。图 9(a) 显示, AUV 有效地维持了表 3 中给出的恒定速度 u 、 v 和 w 。此外, 图 9(b) 的图表描绘了在 x 方向上的持续运动, 同时在 y 和 z 轴上没有明显变化。此外, 图 10 描绘了角速度, 所有角速度均保持为零。AUV 的横滚角稳定在 -2.594° , 而俯仰角保持在 -0.72° 。值得注意的是, 偏航角保持恒定为 0° 。显然, 车辆保持了表 3 中给出的修整值的稳态。

4.3 Nonlinear Controller

为了实现所需的速度、深度和转向, 实施了传统控制器和SMC。非线性模型如公式(14)所示, 并在 MATLAB-Simulink中实现。Simulink中研究车辆的原理图如图11所示。References子系统用作期望值的来源, 包含速度、y位置、z位置和横滚角。这些期望值随后作为输入传递给控制器模块。在控制器模块内, 存在四个不同的控制器, 每个控制器对应一个特定的参考信号。每个控制器

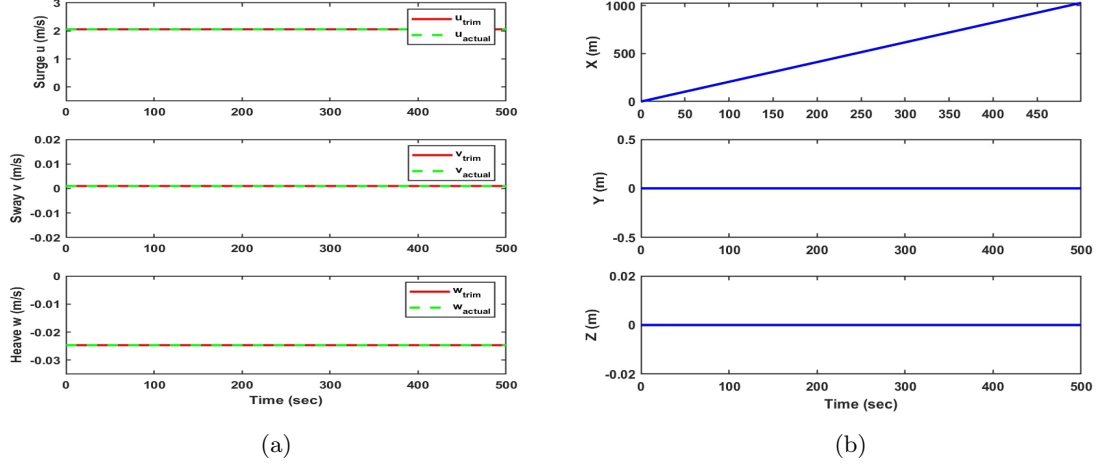


Fig. 9: (a) Body-frame linear velocities u , v and w . (b) AUV motion in the earth-fixed frame.

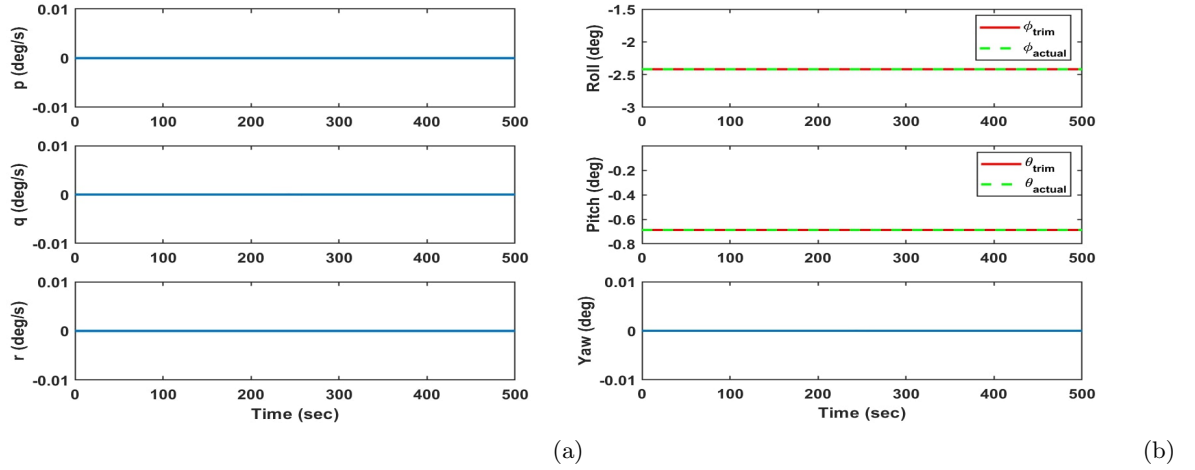


Fig. 10: (a) Body-frame angular velocities u , v and w . (b) Earth-frame angular positions ϕ , θ and ψ .

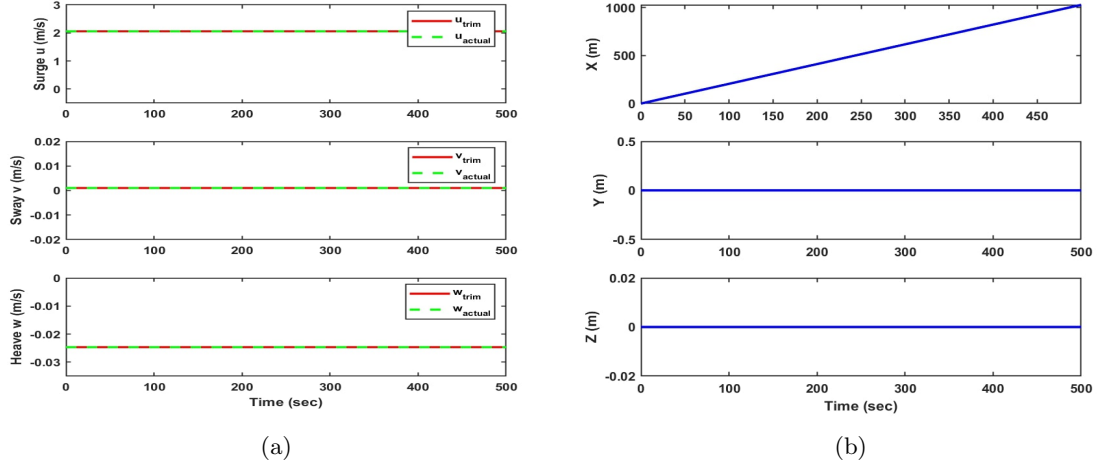


图9: (a) 机体坐标系下的线速度 u , v 和 w 。 (b) AUV 在地面固定坐标系中的运动。

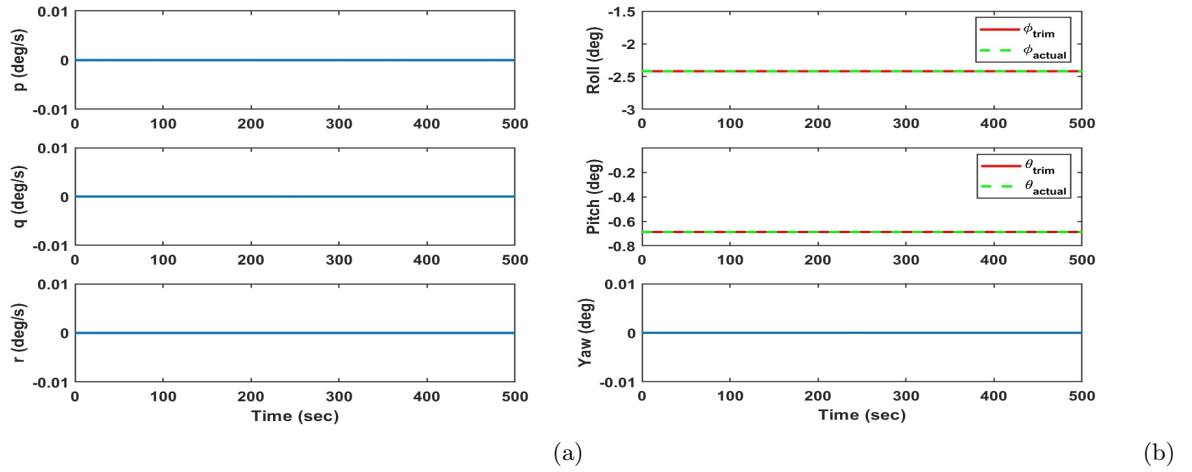


图10: (a) 机体坐标系角速度 u , v 和 w 。 (b) 地球坐标系角位置 ϕ , θ 和 ψ 。

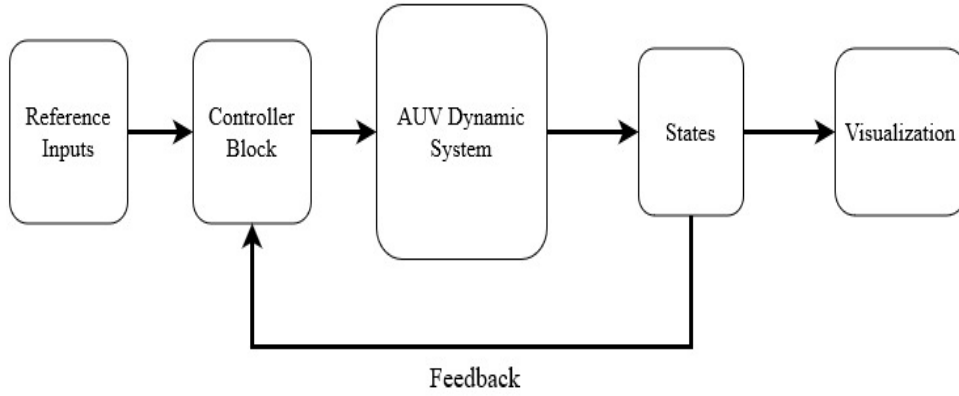


Fig. 11: Block diagram of Nonlinear AUV model

operates as an independent subsystem, employing the principles elucidated in the linear control section. Eventually, it calculates the requisite control commands needed for tracking. The controller block is either conventional controllers or SMC. Further, the AUV dynamics block takes the control inputs furnished by the controller block along with the state feedback information. These combined elements facilitate the estimation of the subsequent state of the AUV.

This simulation model will be used for all our simulations. To demonstrate the efficacy of both conventional control methods and the Sliding Mode Controller (SMC), a series of simulations are conducted. The simulations are structured as follows: Initially, the performance of a conventional controller is assessed. This evaluation encompasses scenarios involving variations in the center of gravity (CG) and the presence of ocean currents. The Conventional controller's response is observed under these conditions to gauge its effectiveness in maintaining stability and control. Subsequently, the focus shifts to the Sliding Mode Controller. The efficacy of SMC is demonstrated using lawn mowing in the presence of CG variations, ocean currents, and parameter variation ($X_{u|u}$ value changed by 50%). By subjecting the SMC to these challenges, its robustness and capability to navigate and counteract disturbances are evaluated.

Conventional Controller: A comparative assessment of the Conventional controller's performance is conducted under two distinct scenarios: with and without variation in the center of gravity (CG) location. This evaluation aims to discern the impact of CG location changes on the controller's effectiveness in regulating the system. In the first scenario, the Conventional controller is examined in a setting where the CG location remains constant. The system's response to input commands and disturbances is analyzed, providing insight into the controller's inherent capabilities in maintaining stability and precision. In the second scenario, the Conventional controller's performance is investigated when confronted with variations in the CG location ($x_{cg} = 8mm$). This dynamic factor introduces an additional layer of complexity, potentially affecting the system's stability and control.

Figure 12 shows a set of simulation data with and without the variation of the location of CG. For this simulation the references used are: $u_{ref} = 4$ knots, $z_{ref} = 60m$, $y_{ref} = 0$ and $\phi_{ref} = 0$. Here first subplot shows the variation of depths in both cases along with reference. The second subplot shows the variation of speed in both cases whereas the third and fourth show the roll and pitch angles respectively for both cases.

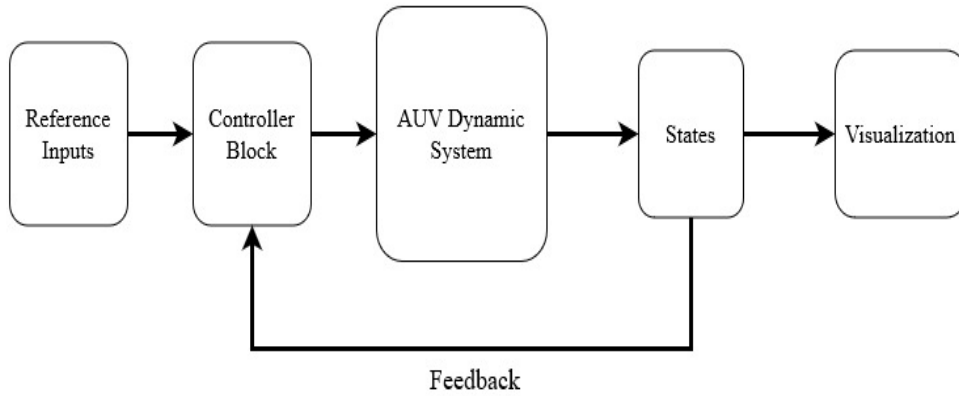


图11: 非线性AUV模型的框图

作为一个独立的子系统运行，采用线性控制部分阐明的原理。最终，它计算跟踪所需的控制指令。控制器模块可以是传统控制器或滑模控制（SMC）。此外，AUV动力学模块接收控制器模块提供的控制输入以及状态反馈信息。这些组合元素有助于估计AUV的下一个状态。

该仿真模型将用于我们所有的仿真。为了展示传统控制方法和滑模控制器（SMC）的有效性，进行了系列仿真。仿真结构如下：首先，评估传统控制器的性能。该评估涵盖重心（CG）变化和洋流存在的场景。在这些条件下观察传统控制器的响应，以评估其在保持稳定性和控制方面的有效性。随后，重点转向滑模控制器。通过在存在CG变化、洋流以及参数变化（ $X_{u|u}$ 值变化50%）的情况下进行割草操作，展示SMC的有效性。通过让SMC面临这些挑战，评估其鲁棒性以及导航和抵御干扰的能力。

Conventional Controller: 在两种不同的情境下，对传统控制器的性能进行了比较评估：有重心(CG)位置变化和无重心(CG)位置变化。此次评估旨在辨别CG位置变化对控制器调节系统有效性的影响。在第一种情境中，传统控制器在CG位置保持不变的情况下进行分析。系统对输入指令和干扰的响应被研究，从而提供关于控制器在维持稳定性和精度方面的固有能力的见解。在第二种情境中，当传统控制器面对CG位置变化($x_{cg} = 8mm$)时，其性能被进一步调查。这个动态因素引入了额外的复杂性，可能影响系统的稳定性和控制效果。

图12显示了一组具有和不具有重心位置变化的仿真数据。对于此仿真使用的参考值为： $u_{ref} = 4$ 节， $z_{ref} = 60$ 米， $y_{ref} = 0$ 和 $\phi_{ref} = 0$ 。第一个子图显示了在两种情况下深度的变化以及参考值。第二个子图显示了两种情况下的速度变化，而第三个和第四个子图分别显示了两种情况下的横摇角和纵俯角。

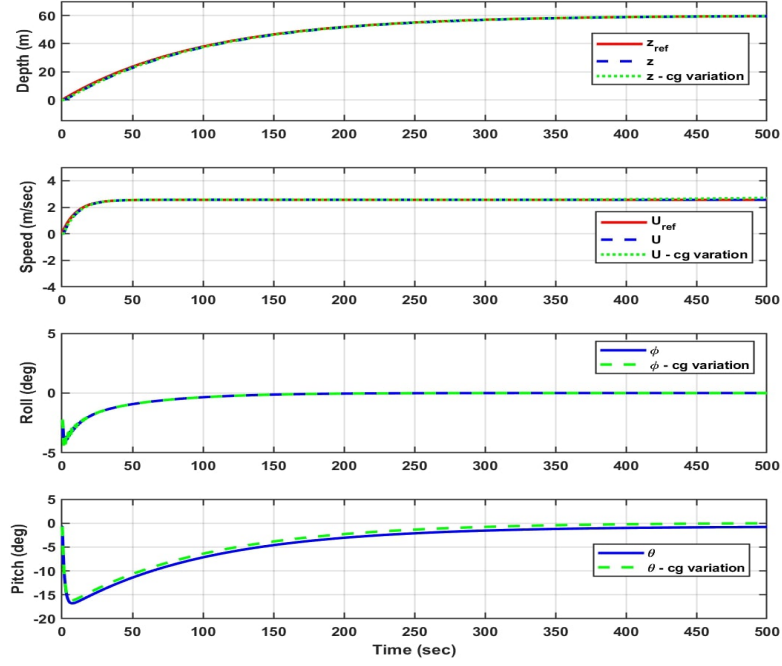


Fig. 12: Comparison of Conventional controller performance with and without variation in CG location

In order to effectively nullify the roll angle ϕ , a strategic approach is employed as outlined below:

$$\begin{aligned}
 \delta_{s1} &= \delta_s + \kappa \frac{\delta_a}{2} \\
 \delta_{s2} &= \delta_s - \kappa \frac{\delta_a}{2} \\
 \delta_{r1} &= \delta_r + (1 - \kappa) \frac{\delta_a}{2} \\
 \delta_{r2} &= \delta_r - (1 - \kappa) \frac{\delta_a}{2}
 \end{aligned} \tag{71}$$

where, the parameter κ assumes a pivotal role, determining the degree of emphasis attributed to the stern and rudder angles within the endeavor to nullify the roll angle and δ_a represents the requisite output emerging from the roll controller housed within the controller block, as depicted in Figure 11. This orchestrated strategy meticulously adjusts control inputs through the defined equations, aiming to achieve the specific objective of nullifying the roll angle. This is evidently shown in Figure 12. The vehicle's roll angle ϕ gradually approaches zero from its initial value of -2.59° . While a minor undershoot is observed during the initial phase, the roll angle eventually stabilizes at zero, demonstrating a successful convergence to the desired state. Analyzing these outcomes collectively reveals that even with minor adjustments in x_{cg} , the system adeptly achieves

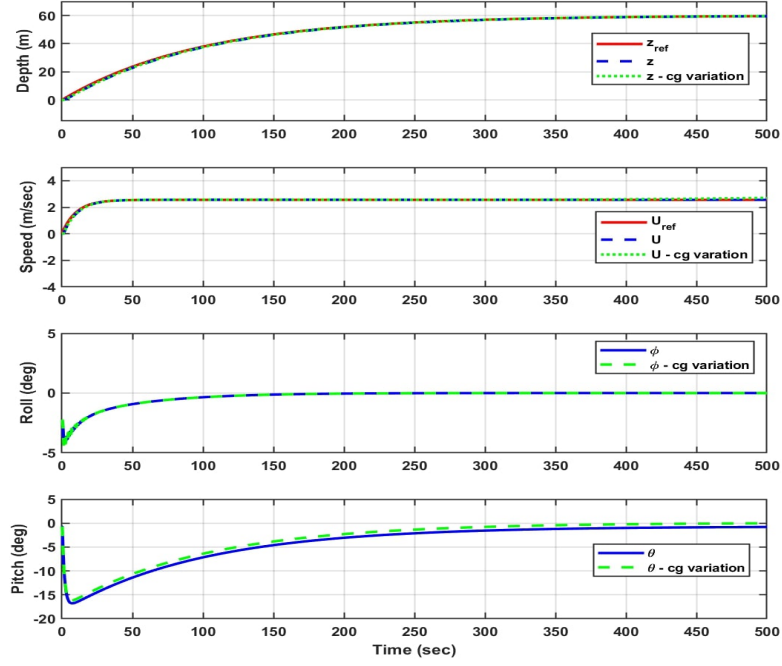


图12: 传统控制器在有无CG位置变化情况下的性能比较

为了有效抵消滚转角 ϕ , 采用如下所述的战略方法:

$$\begin{aligned}
 \delta_{s1} &= \delta_s + \kappa \frac{\delta_a}{2} \\
 \delta_{s2} &= \delta_s - \kappa \frac{\delta_a}{2} \\
 \delta_{r1} &= \delta_r + (1 - \kappa) \frac{\delta_a}{2} \\
 \delta_{r2} &= \delta_r - (1 - \kappa) \frac{\delta_a}{2}
 \end{aligned} \tag{71}$$

其中, 参数 κ 扮演关键角色, 决定在消除滚转角的过程中舵和舵角所赋予的强调程度, 而 δ_a 表示从控制器块中的滚转控制器所产生的所需输出, 如图 11 所示。这一精心安排的策略通过定义的方程仔细调整控制输入, 旨在实现消除滚转角的特定目标。如图 12 所示, 车辆的滚转角 ϕ 从初始值 -2.59° 逐渐趋近于零。虽然在初始阶段观察到轻微的下冲, 但滚转角最终稳定在零, 表明成功收敛到期望状态。综合分析这些结果表明, 即使在 x_{cg} 上进行轻微调整, 系统也能熟练实现

the desired outcomes. It is noteworthy that slight alterations in trim values for specific variables might occur. By contrasting the results with the first scenario, the Conventional controller's robustness and adaptability in handling CG location changes can be elucidated.

Further, we implement the ocean current model into the simulation framework. The ocean current model considered is simulated using eq. (33). The ocean current relative velocity, V_c , is generated with Gaussian input, $\zeta = 1$ and initial condition $V_{c0} = 0.5$. We have set α_c to $0.035rad$ (2°) and β_c to $1.221rad$ (70°). This ocean current model mostly affects the speed and depth of the AUV. This model of ocean currents was added to the same simulations as performed earlier and results are shown in Figure 13.

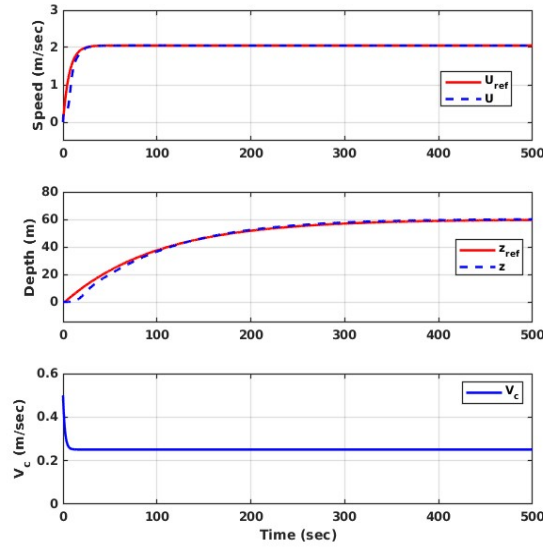


Fig. 13: Simulation results with ocean currents

The first subplot reveals a minute difference between the desired speed U_{ref} and the actual speed of the vehicle U . Although there is a slight transient and persists for very few seconds. Later the controller seems to be successful in maintaining the desired speed. Similarly, the depth z in the second subplot follows the desired depth command z_{ref} . The third subplot shows the variation of U_c over time. It can be seen that U_c begins with the initial value $0.5m/s$ and reaches a steady state value of $0.25m/s$. The simulations have showcased the effectiveness of conventional controllers in maintaining desired vehicle trajectories and depths even in the presence of ocean currents. More complex ocean models can be used to increase the accuracy of the simulation.

Comparison of SMC and Conventional controller: Lawn Mowing Manoeuvre In this section, we build a control system architecture where sliding mode controllers are used for the inner pitch and yaw control loops. The core idea behind sliding mode control is to shape a certain trajectory of system states as they advance over a defined manifold known as the sliding surface. Usually, the sliding surface depends on the tracking error (e). For inner loop pitch control θ is the state variable

期望的结果。值得注意的是，特定变量的微调值可能会出现轻微变化。通过将结果与第一个情境进行对比，可以阐明传统控制器在处理CG位置变化时的鲁棒性和适应性。

此外，我们将海流模型实现到仿真框架中。所考虑的海流模型是使用公式 (33) 进行模拟的。海流相对速度 V_c 是通过高斯输入 $\zeta = 1$ 和初始条件 $V_{c0} = 0.5$ 生成的。我们已将 α_c 设置为 $0.035rad$ (2°)，并将 β_c 设置为 $1.221rad$ (70°)。该海流模型主要影响 AUV 的速度和深度。这个海流模型被添加到之前执行的相同仿真中，结果如图 13 所示。

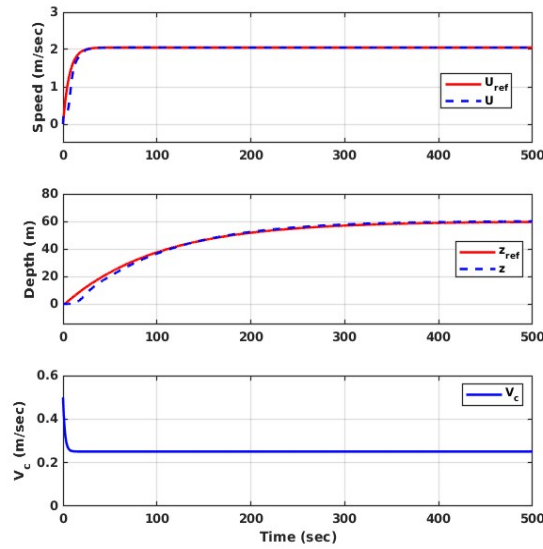


图13: 含海洋洋流的仿真结果

第一个子图显示了期望速度 U_{ref} 与车辆实际速度 U 之间的微小差异。虽然存在轻微的瞬态，并且持续时间非常短。随后，控制器似乎成功地维持了期望速度。类似地，第二个子图中的深度 z 遵循期望深度指令 z_{ref} 。第三个子图显示了 U_c 随时间的变化。可以看到 U_c 从初始值 $0.5m/s$ 开始，并达到稳态值 $0.25m/s$ 。模拟结果展示了传统控制器在保持期望车辆轨迹和深度方面的有效性，即使在存在海流的情况下。可以使用更复杂的海洋模型来提高模拟的精确性。

Comparison of SMC and Conventional controller: Lawn Mowing Manoeuvre 在本节中，我们构建了一个控制系统架构，其中滑模控制器用于内环俯仰和偏航控制回路。滑模控制的核心思想是将系统状态随时间推进的一定轨迹塑造成一个称为滑动面 (sliding surface) 的定义流形。通常，滑动面依赖于跟踪误差 (e)。对于内环俯仰控制， θ 是状态变量

and for inner loop yaw control ψ is the state variable. The sliding surfaces for pitch and yaw inner loops respectively are,

$$s_{pitch} = \gamma_{\theta_1} \dot{e}_\theta + \gamma_{\theta_2} e_\theta \quad (72)$$

$$s_{yaw} = \gamma_{\psi_1} \dot{e}_\psi + \gamma_{\psi_2} e_\psi \quad (73)$$

where e_θ is the error between θ_{ref} and θ and e_ψ error between ψ_{ref} and ψ . The sliding mode controller design process involves tuning these constants γ_{θ_1} , γ_{θ_2} , γ_{ψ_1} and γ_{ψ_2} such that the derivatives of respective control surfaces go to zero.

$$\dot{s}_{pitch} = 0 \quad (74)$$

$$\dot{s}_{yaw} = 0 \quad (75)$$

By actively steering the system states toward and subsequently maintaining them on the sliding surface, sliding mode control offers a robust and dependable strategy well-suited to a spectrum of intricate and dynamically evolving systems. The control law for the inner pitch loop

$$u_{pitch} = \delta_s = \gamma_{\theta_3} \text{sgn}(s_{pitch}) \quad (76)$$

and for the yaw is given by,

$$u_{yaw} = \delta_r = \gamma_{\psi_3} \text{sgn}(s_{psi}) \quad (77)$$

The control law in SMC is designed to drive the system's states to the sliding surface and maintain them there. However, a critical challenge in SMC is the occurrence of chattering, which refers to rapid and frequent switching between control actions near the sliding surface. Chattering can lead to undesirable high-frequency oscillations in the control signal, potentially affecting system performance and wear on actuators. To mitigate this, we use the saturation function in the control law. These functions can limit the rate of change of control inputs, preventing rapid fluctuations and reducing chattering.

A classical lawn mowing manoeuvre is performed to showcase the ability of conventional and SMC controllers. This manoeuvre is essentially used for underwater object detection, search and rescue, environmental surveys and many other applications. It gives a broad sense of the ability of the vehicle to operate autonomously. The conventional controller architecture remains unchanged, serving as a benchmark for evaluation. In contrast, SMC system is introduced to handle the inner pitch and yaw control functions, following the equations eq. (72), eq. (73), eq. (76), and eq. (77). A visual representation of the AUV's positional dynamics (in XY plane) is vividly presented through the plot in Figure 14. Evidently, both controllers successfully executed the planned manoeuvre.

Further, the comparative analysis between the two controllers is conducted under the umbrella of uncertainties. In this experimental framework, the target speed is set at $U_{ref} = 4 \text{ Knots}$ and the vehicle is assumed to be at the desired depth of $60m$. To comprehend the system's behaviour in response to uncertainties, we subject the system to a series of desired commands that yield a lawn-mowing manoeuvre. Notably, this examination involves varying a single parameter at a time, allowing us to scrutinize the performance of both controllers. During this investigation, our focus was the error along the y-axis as there was minimal or no change in the x-direction, hence the root-mean-square error (RMSE) in the y-direction is recorded for each variation as

$$RMSE = \sqrt{\frac{(y_{ref} - y)^2}{N}}, \quad (78)$$

where, y_{ref} is the desired value in the y-direction, y is the actual value and N is the number of the samples. This choice serves as a metric to assess the effectiveness of the controllers under the

对于内环偏航控制, ψ 是状态变量。俯仰和偏航内环的滑动面分别为,

$$s_{pitch} = \gamma_{\theta_1} \dot{e}_\theta + \gamma_{\theta_2} e_\theta \quad (72)$$

$$s_{yaw} = \gamma_{\psi_1} \dot{e}_\psi + \gamma_{\psi_2} e_\psi \quad (73)$$

其中 e_θ 是 θ_{ref} 与 θ 之间的误差, e_ψ 是 ψ_{ref} 与 ψ 之间的误差。滑模控制器的设计过程涉及调整这些常数 γ_{θ_1} 、 γ_{θ_2} 、 γ_{ψ_1} 和 γ_{ψ_2} , 以使相应控制面的导数趋于零。

$$\dot{s}_{pitch} = 0 \quad (74)$$

$$\dot{s}_{yaw} = 0 \quad (75)$$

通过主动将系统状态引导到滑动面上并随后保持在滑动面上, 滑模控制提供了一种稳健且可靠的策略, 非常适合一系列复杂且动态变化的系统。内俯仰环的控制律

$$u_{pitch} = \delta_s = \gamma_{\theta_3} \text{sgn}(s_{pitch}) \quad (76)$$

对于偏航角, 其表示为,

$$u_{yaw} = \delta_r = \gamma_{\psi_3} \text{sgn}(s_{psi}) \quad (77)$$

SMC中的控制规律旨在将系统状态驱动到滑模面并将其保持在该面上。然而, SMC中的一个关键挑战是颤振的发生, 颤振是指在滑模面附近控制动作的快速频繁切换。颤振可能导致控制信号中出现不希望的高频振荡, 可能影响系统性能并增加执行器的磨损。为减轻这一问题, 我们在控制规律中使用饱和函数。这些函数可以限制控制输入的变化速率, 防止快速波动并减少颤振。

执行了一项经典的割草机操作以展示传统控制器和SMC控制器的能力。这一操作本质上用于水下物体检测、搜救、环境调查以及许多其他应用。它提供了车辆自主操作能力的总体概念。传统控制器架构保持不变, 作为评估的基准。相比之下, 引入SMC系统来处理内俯仰和偏航控制功能, 遵循方程式 eq. (72)、eq. (73)、eq. (76) 和 eq. (77)。通过图14中的绘图, 生动展示了AUV在XY平面上的位置动态。显然, 两种控制器都成功执行了预定的操作。

此外, 在不确定性条件下对两种控制器进行了比较分析。在这个实验框架中, 目标速度设定为 $U_{ref} = 4 \text{ Knots}$, 车辆假定处于期望深度 $60m$ 。为了理解系统对不确定性的响应行为, 我们对系统施加一系列产生杂草修剪 (lawn-mowing) 动作的期望指令。值得注意的是, 这一研究涉及一次只改变一个参数, 从而使我们能够仔细检查两种控制器的性能。在这一研究过程中, 我们关注的是 y 轴方向的误差, 因为 x 方向几乎没有变化, 因此记录了每次变化的 y 方向均方根误差 (RMSE)。

$$RMSE = \sqrt{\frac{(y_{ref} - y)^2}{N}}, \quad (78)$$

其中, y_{ref} 是 y 方向上的期望值, y 是实际值, N 是样本数量。此选择作为评估控制器有效性的度量指标。

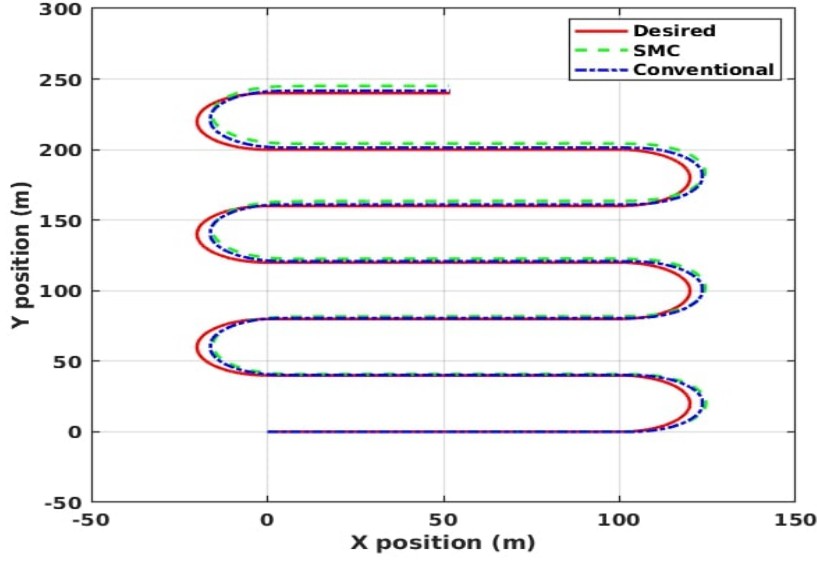


Fig. 14: Lawn mowing in XY plane with SMC and Conventional controller

influence of parameter variations. To ensure clarity and precision, only one parameter is altered and results are recorded. The comparison of two controllers in terms of (RMSE) are tabulated in Table 4. Similarly, the simulations are repeated for different values of buoyancy. The results are given in 5.

Table 4: RMSE of Conventional and SMC due to variation in hydrodynamic coefficients

Controller	Coefficient								
	$X_{u u }$			M_{uw}			Z_{uw}		
% change	-25%	0%	+25%	-25%	0%	+25%	-25%	0%	+25%
Conventional	0.506	0.504	0.504	0.502	0.504	0.507	0.503	0.504	0.507
SMC	1.600	1.614	1.623	1.613	1.614	1.611	1.600	1.614	1.611

As observed from Table 4 and 5, the error in the case of conventional is found to be lesser compared to SMC controller. When observed closely, in the case of SMC it can be seen that with parameter variation the change in error is minimal compared to conventional controller.

Further, the comparison of the two strategies is shown in Figure 15. The simulations encompass a simultaneous variation in two parameters, including variations in the Center of Gravity (CG) location ($x_{cg} = 8mm$) and a 50% alteration in the axial drag coefficient $X_{u|u|}$. Additionally, a consistent external disturbance is introduced in the form of a constant ocean current, characterized by a relative velocity of $U_c = 0.1 m/s$, an angle of attack $\alpha_c = 0^\circ$, and a side-slip angle $\beta_c = 20^\circ$. Figure 15 shows the paths followed in the XY plane and highlights the difference between the desired

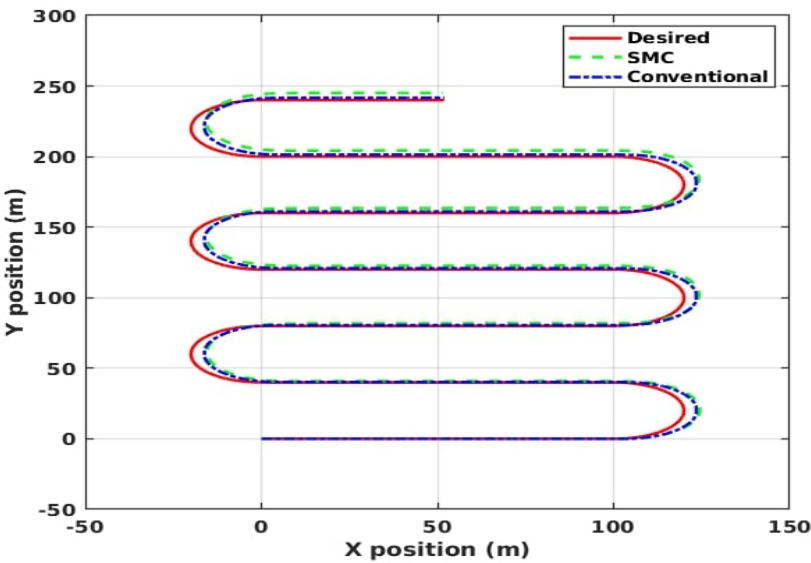


图14: 在XY平面上使用SMC和传统控制器进行草坪割草

参数变化的影响。为了确保清晰和精确，只改变一个参数并记录结果。两个控制器在均方根误差（RMSE）方面的比较列在表4中。同样，不同浮力值下的仿真也被重复进行。结果见表5。

表 4: RMSE o 由于液动力变化的常规和SMC							IC 系数		
Controller	Coeffecient								
	$X_{u u }$			M_{uw}			Z_{uw}		
% change	-25%	0%	+25%	-25%	0%	+25%	-25%	0%	+25%
Conventional	0.506	0.504	0.504	0.502	0.504	0.507	0.503	0.504	0.507
SMC	1.600	1.614	1.623	1.613	1.614	1.611	1.600	1.614	1.611

从表4和表5可以看出，传统控制器的误差相比SMC控制器来说较小。仔细观察时，可以发现对于SMC控制器，随着参数的变化，误差的变化相比传统控制器是最小的。

此外，两个策略的比较如图15所示。模拟包括两个参数的同时变化，包括重心（CG）位置的变化（ $x_{cg} = 8mm$ ）以及轴向阻力系数 $X_{u|u|}$ 的50%变化。此外，引入了稳定的外部扰动，即恒定的洋流，其特征为相对速度 $U_c = 0.1\text{ m/s}$ 、攻角 $\alpha_c = 0^\circ$ 以及侧滑角 $\beta_c = 20^\circ$ 。图15显示了在 XY 平面上的路径，并突出了期望路径之间的差异

Table 5: RMSE of Conventional and SMC due to variation in buoyancy

Controller	(B-W)/g (grams)			
	100	340	650	1000
Conventional	0.513	0.505	0.495	0.483
SMC	1.613	1.614	1.612	1.613

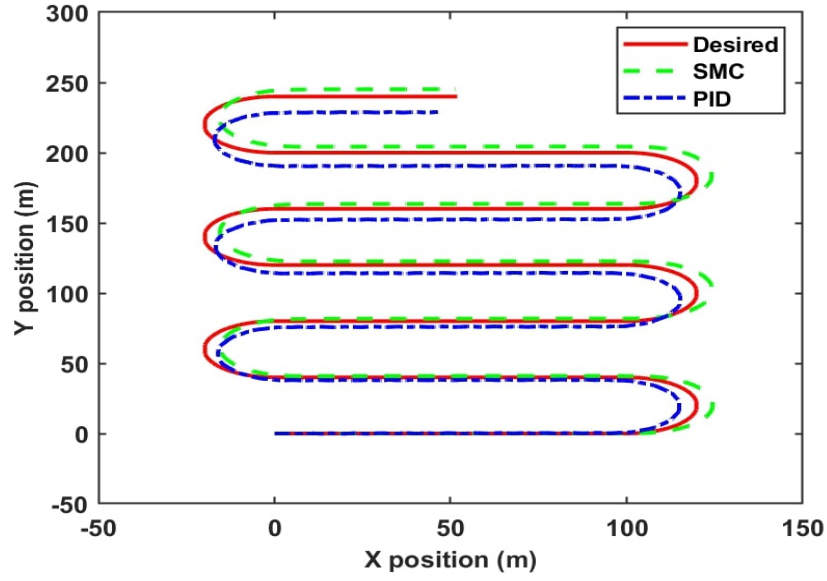
Fig. 15: Lawn mowing in XY plane of SMC and Conventional controller with parameter variations

表5: 由于浮力变化, 传统方法和SMC的均方根误差 (RMSE)

Controller	(B-W)/g (grams)			
	100	340	650	1000
Conventional	0.513	0.505	0.495	0.483
SMC	1.613	1.614	1.612	1.613

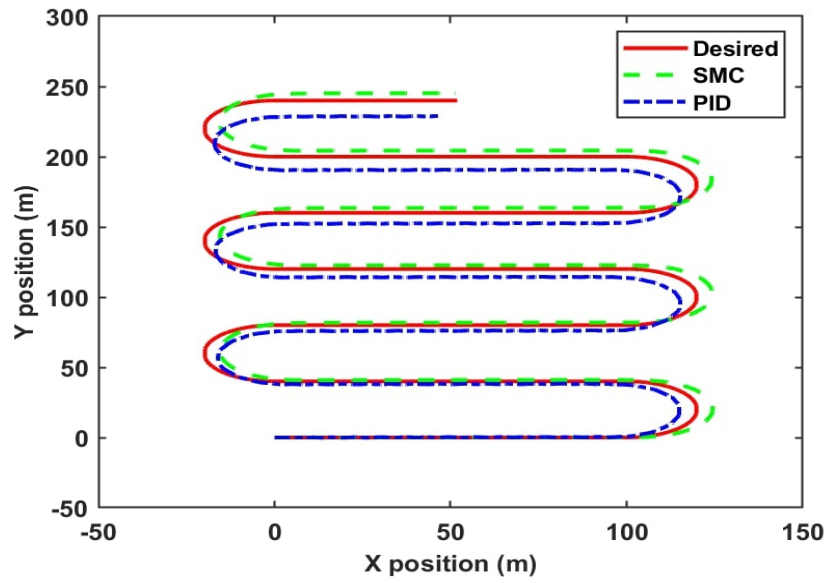


图15: 在参数变化下 SMC 和常规控制器的 XY 平面割草情况

path (shown in solid red) and the trajectories tracked with the conventional controller and SMC. Evidently, there's a noticeable deviation with the conventional controller, especially from the second cycle starting at 100m on the Y-axis. On the other hand, the trajectory controlled by the SMC strategy remains remarkably steady, with minimal deviations from the intended path. These results strongly emphasize the robustness of the Sliding Mode Control (SMC) approach. It counteracts challenges posed by ocean currents, parameter variations, and shifts in the Center of Gravity (CG). The RMSE values are found to be 3.694 and 1.632 for conventional and SMC controllers respectively. It is evident that error with conventional controller has increased significantly compared to SMC. In conclusion, the SMC approach emerges as a reliable controller even in complex and changing underwater conditions.

5 Conclusions

In conclusion, this book chapter undertook an in-depth exploration of the modelling and robust control of an Autonomous Underwater Vehicle amidst the presence of uncertainties. The dynamic behaviour of the vehicle was represented with a 6-DoF model, accounting for the impact of ocean currents. The level flight requirements are derived across various speeds, and eventually 6-DoF model was dissected into horizontal and vertical subsystems, facilitating comprehensive linear analysis.

This study investigated different linear controllers, including depth, yaw, and speed subsystems. Simulation results were showcased, particularly highlighting the efficacy of these controllers in varying operational contexts, including level flight at a speed of 4 knots. Moreover, nonlinear control methodologies, encompassing both conventional and sliding-mode control were scrutinized. The controller design process diligently addressed a spectrum of uncertainties, ranging from ocean currents to parameter fluctuations, Center of Gravity (CG) deviations, and buoyancy variations. Importantly, a comparative assessment between the outcomes of conventional and SMC controllers was presented with a lawn-mowing manoeuvre scenario. This revealed SMC's enhanced robustness against disturbances and parameter uncertainties, albeit with the added complexity inherent to its design and implementation.

In essence, this book chapter delivered an encompassing study, elucidating the intricate interplay between modelling, control strategies, uncertainties, and robustness in the realm of AUVs. By delving into both linear and nonlinear approaches, the chapter provided valuable insights into the trade-offs between robustness and complexity, contributing to a holistic understanding of AUV control and navigation in challenging underwater environments.

Appendix

The myring hull parameters, force and moment coefficients of the research vehicle [36], [57] considered in the study are given in Table 6 and Table 7 respectively.

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路径（以实红色显示）以及使用传统控制器和滑模控制（SMC）跟踪的轨迹。显然，传统控制器存在明显偏差，尤其是在从 Y 轴 $100m$ 开始的第二个周期。另一方面，由 SMC 策略控制的轨迹保持非常稳定，与预期路径的偏差最小。这些结果强烈强调了滑模控制（SMC）方法的稳健性。它可以应对洋流、参数变化和重心（CG）移动带来的挑战。均方根误差（RMSE）值分别为传统控制器和 SMC 控制器为 3.694 和 1.632。显然，相比于 SMC，传统控制器的误差显著增加。总之，即使在复杂多变的水下环境中，SMC 方法也表现出可靠的控制能力。

5 Conclusions

总之，本书章节对自主水下航行器在存在不确定性的情况下的建模与鲁棒控制进行了深入探索。采用六自由度（6-DoF）模型表示了车辆的动态行为，并考虑了洋流的影响。根据不同速度推导了平飞要求，最终将六自由度模型拆分为水平和垂直子系统，从而便于进行全面的线性分析。

本研究调查了不同的线性控制器，包括深度、偏航和速度子系统。展示了仿真结果，特别强调了这些控制器在不同操作环境下的有效性，包括以4节速度进行平飞。此外，还审查了非线性控制方法，包括传统控制和滑模控制。控制器设计过程仔细考虑了一系列不确定性，从洋流到参数波动、重心（CG）偏移以及浮力变化。重要的是，研究还在割草机机动场景下对传统控制器和滑模控制器的结果进行了比较评估。这显示了滑模控制器在应对干扰和参数不确定性方面的增强鲁棒性，尽管其设计和实现具有固有的复杂性。

本质上，本书章节提供了一项全面的研究，阐明了自主水下航行器（AUV）领域中建模、控制策略、不确定性与鲁棒性之间的复杂相互作用。通过探讨线性和非线性方法，该章节为鲁棒性与复杂性之间的权衡提供了宝贵的见解，有助于全面理解在复杂水下环境中AUV的控制和导航。

Appendix

本研究中考虑的研究车辆[36]、[57]的梅林船体参数、力和力矩系数分别列于表6和表7。

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Table 6: Myring Parameters for AUV

Description	Unit	Value
Nose Length (a)	m	0.191
Nose Offset (a_{offset})	m	0.0165
Mid-section Length (b)	m	0.654
Tail section Length (c)	m	0.541
Tail Offset (c_{offset})	m	0.0368
Exponential Coefficient (n)	-	2
Included angle at tail (θ)	rad	0.436
Max Hull Diameter (d)	m	0.191
Vehicle Forward Length (l_f)	m	0.828
Total Vehicle Length (l)	m	1.33

Table 7: Force and Moment coefficients of research vehicle

Force Parameter	Unit	Value	Moment Parameter	Unit	Value
X_{uu}	kg/m	-1.62	K_{pp}	$kg.m^2/rad^2$	-0.001
$X_{\dot{u}}$	kg	-0.93	$K_{\dot{p}}$	$kg.m^2/rad$	-0.014
X_{wq}	kg/rad	-35.5	M_{ww}	kg	3.18
X_{qq}	$kg.m/rad$	1.93	M_{qq}	$kg.m^2/rad^2$	-9.4
X_{vr}	kg/rad	35.5	$M_{\dot{w}}$	$kg.m$	-1.93
X_{rr}	$kg.m/rad$	-1.93	M_{uw}	kg	24
Y_{vv}	kg/m	-131	$M_{\dot{q}}$	$kg.m^2/rad$	-4.88
Y_{rr}	$kg.m/rad^2$	0.632	M_{uq}	$kg.m/rad$	-2
Y_{uv}	kg/m	-28.6	M_{vp}	$kg.m/rad$	-1.93
$Y_{\dot{v}}$	kg	-35.5	M_{rp}	$kg.m^2/rad^2$	4.86
$Y_{\dot{r}}$	$kg.m/rad$	1.93	$M_{uu\delta_s}$	kg/rad	-6.15
Y_{ur}	kg/rad	5.22	N_{vv}	kg	-3.18
Y_{wp}	kg/rad	35.5	N_{rr}	$kg.m^2/rad^2$	-9.4
Y_{pq}	$kg.m/rad$	1.93	N_{uv}	kg	-24
$Y_{uu\delta_r}$	$kg/(m.rad)$	9.64	$N_{\dot{v}}$	$kg.m$	1.93
Z_{ww}	kg/m	-131	$N_{\dot{r}}$	$kg.m^2/rad^2$	-4.88
Z_{qq}	$kg.m/rad^2$	-0.632	N_{ur}	$kg.m/rad$	-2
Z_{uw}	kg/m	-28.6	N_{wp}	$kg.m/rad$	-1.93
$Z_{\dot{w}}$	kg	-35.5	N_{pq}	$kg.m^2/rad^2$	-4.86
$Z_{\dot{q}}$	$kg.m/rad$	-1.93	$N_{uu\delta_r}$	kg/rad	-6.15
Z_{uq}	kg/rad	-5.22	-	-	-
Z_{vp}	kg/rad	-35.5	-	-	-
Z_{rp}	kg/rad	1.93	-	-	-
$Z_{uu\delta_s}$	$kg/(m.rad)$	-9.64	-	-	-

表6: AUV的鼓膜参数

Description	Unit	Value
Nose Length (a)	m	0.191
Nose Offset (a_{offset})	m	0.0165
Mid-section Length (b)	m	0.654
Tail section Length (c)	m	0.541
Tail Offset (c_{offset})	m	0.0368
Exponential Coefficient (n)	-	2
Included angle at tail (θ)	rad	0.436
Max Hull Diameter (d)	m	0.191
Vehicle Forward Length (l_f)	m	0.828
Total Vehicle Length (l)	m	1.33

表 7: 研究车辆的力和力矩系数

Force Parameter	Unit	Value	Moment Parameter	Unit	Value
X_{uu}	kg/m	-1.62	K_{pp}	$kg.m^2/rad^2$	-0.001
$X_{\dot{u}}$	kg	-0.93	$K_{\dot{p}}$	$kg.m^2/rad$	-0.014
X_{wq}	kg/rad	-35.5	M_{ww}	kg	3.18
X_{qq}	$kg.m/rad$	1.93	M_{qq}	$kg.m^2/rad^2$	-9.4
X_{vr}	kg/rad	35.5	$M_{\dot{w}}$	$kg.m$	-1.93
X_{rr}	$kg.m/rad$	-1.93	M_{uw}	kg	24
Y_{vv}	kg/m	-131	$M_{\dot{q}}$	$kg.m^2/rad$	-4.88
Y_{rr}	$kg.m/rad^2$	0.632	M_{uq}	$kg.m/rad$	-2
Y_{uv}	kg/m	-28.6	M_{vp}	$kg.m/rad$	-1.93
$Y_{\dot{v}}$	kg	-35.5	M_{rp}	$kg.m^2/rad^2$	4.86
$Y_{\dot{r}}$	$kg.m/rad$	1.93	$M_{uu\delta_s}$	kg/rad	-6.15
Y_{ur}	kg/rad	5.22	N_{vv}	kg	-3.18
Y_{wp}	kg/rad	35.5	N_{rr}	$kg.m^2/rad^2$	-9.4
Y_{pq}	$kg.m/rad$	1.93	N_{uv}	kg	-24
$Y_{uu\delta_r}$	$kg/(m.rad)$	9.64	$N_{\dot{v}}$	$kg.m$	1.93
Z_{ww}	kg/m	-131	$N_{\dot{r}}$	$kg.m^2/rad^2$	-4.88
Z_{qq}	$kg.m/rad^2$	-0.632	N_{ur}	$kg.m/rad$	-2
Z_{uw}	kg/m	-28.6	N_{wp}	$kg.m/rad$	-1.93
$Z_{\dot{w}}$	kg	-35.5	N_{pq}	$kg.m^2/rad^2$	-4.86
$Z_{\dot{q}}$	$kg.m/rad$	-1.93	$N_{uu\delta_r}$	kg/rad	-6.15
Z_{uq}	kg/rad	-5.22	-	-	-
Z_{vp}	kg/rad	-35.5	-	-	-
Z_{rp}	kg/rad	1.93	-	-	-
$Z_{uu\delta_s}$	$kg/(m.rad)$	-9.64	-	-	-

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